

Electroweak Vacuum Fixation and Residual-Spectrum Rigidity

Gauge quotient, electromagnetic stabilizer, Goldstone translation,
and the one-doublet electroweak normal form

SYM-8 of the Standard Model Symmetry Arc

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Principal result

Fix the anomaly-neutral one-Higgs-doublet electroweak target D_{EW}^{1H} , a selected nonzero fixation branch, nonzero electroweak couplings, the proper local-gauge quotient, and the explicit perturbative spectral premise bundle. Proper local gauge transformations identify gauge-related Higgs representatives rather than generating distinct physical vacua. The connected annihilator of the fixation class is one-dimensional and generated, up to normalization, by

$$Q = T_3 + \frac{1}{2y_H}Y, \quad YH = y_H H.$$

The Higgs-induced gauge-response form is positive semidefinite of rank three, with field-coordinate kernel $C_g^{-1}(\mathbb{R}Q)$. In canonical neutral coordinates this gives one electromagnetic null direction and one nonzero Z-response direction, while the charged directions are $W^\pm$. The same generator reconstructs the inherited electric-charge ratios; proper local electromagnetic operations remain quotient-trivial, whereas a nontrivial boundary or asymptotic action requires an actual surface-charge certificate. The admitted perturbative physical spectrum, the high-energy Goldstone–longitudinal translation, the Higgs bridge/fixation carrier, the custodial interface, and the phase signature are then assembled into one presentation-covariant electroweak normal form.

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Proof order	Framework-neutral QFT reconstruction; quotient and observable certification; AASC role assignment; scope-preserving rival analysis
Direct SYM imports	SYM-0, SYM-1, SYM-3, SYM-5, SYM-6, with the fixed target boundary inherited through SYM-7
Structural imports	Higgs Structural Endpoint Arc; gauge–vacuum compatibility; residual-sector normal form; Abelian charge-ratio rigidity; Standard Model structural spine; realized physical interior
Successor boundary	Faithful global form and extended operators to SYM-9; whole-arc candidate completeness and exhaustion to SYM-11

Abstract

The standard phrase “electroweak gauge symmetry is spontaneously broken” compresses several logically different statements. A gauge-fixed Higgs representative is selected; the gauge-boson response form develops three nonzero directions; one neutral direction remains massless; electric charge acts on physical sectors; Goldstone coordinates can replace longitudinal vector legs in a controlled high-energy amplitude theorem; and a custodial limit can organize selected response relations. Those statements do not share one quotient status, one acting object, or one structural role. This paper constructs SYM-8 of the Standard Model Symmetry Arc by separating them and then recomposing them into a fixed-target electroweak normal form.

For the anomaly-neutral one-Higgs-doublet target with nonzero fixation branch, proper local gauge transformations identify gauge-related Higgs representatives and prevent a gauge-variant local expectation value from serving as an ordinary physical order parameter for local gauge redundancy. The connected annihilator of any representative is one-dimensional and generated, up to normalization, by

$$Q = T_3 + \frac{1}{2y_H}Y, \quad YH = y_H H.$$

In the convention $y_H = \frac{1}{2}$, this is $Q = T_3 + Y$. The positive-semidefinite gauge-response form induced by the Higgs kinetic term has rank three and field-coordinate kernel

$$\mathbb{R}Q_g = C_g^{-1}(\mathbb{R}Q),$$

where C_g inserts the nonzero electroweak couplings. In a canonically normalized neutral basis, the null and image directions are the electromagnetic and Z-response directions; the charged directions are $W^\pm$. The generator Q , linked to the null field direction by C_g and the current pairing, reconstructs the inherited quark, lepton, Higgs, and vector-boson charge ratios. Simultaneous hypercharge and coupling rescaling leaves the interaction pairing and all physical ratios invariant.

Residual electromagnetism is typed by locus. Proper local electromagnetic operations remain quotient-trivial gauge transformations, whereas global or asymptotic transformations can act nontrivially only when a differentiable, integrable, nonzero surface-charge certificate is present. Physical spectrum claims are stated through BRST cohomology, gauge-invariant correlators, physical amplitudes, and gauge-independent real or complex pole classes. Their existence is carried explicitly by the target spectral bundle rather than inferred from response rank alone. Gauge-fixed Goldstone variables are retained as representatives in the longitudinal construction, with the actual high-energy relation

$$\mathcal{A}(V_L^{a_1}, \dots, V_L^{a_n}; X) = \left(\prod_k C_{a_k} \right) \mathcal{A}(\phi^{a_1}, \dots, \phi^{a_n}; X) + \mathcal{O}(m_V/E),$$

not as an exact all-energy field identity or an additional physical-particle count.

Only after these QFT statements are fixed is the AASC role map applied. The Higgs field coordinate, vev, potential chart, gauge choice, neutral basis, and Goldstone chart remain licensed representatives or downstream response coordinates; none is promoted to an independent Higgs identity or residual-sector selector. Custodial $SU(2)_V$ is imported as a separate global or limit structure and is not identified with $U(1)_{\text{em}}$. The integrated theorem proves that the quotient fixation class, connected stabilizer conjugacy class, response rank and coupling-pullback kernel, charge-ratio ledger, residual-action locus split, certificate-conditional spectrum, Goldstone translation, Higgs-carrier status, custodial status, and phase signature are invariant under the admitted gauge, basis, normalization, gauge-fixing, and invertible Higgs-repackaging transformations. A finite fifty-schema rival registry accompanies the paper as an auditable stress test. It records theorem-backed dispositions for the declared examples, but it is not promoted into a semantic proof that every possible same-target rival has been generated. Candidate completeness and whole-arc exhaustion remain the explicit burden of SYM-11.

Keywords: electroweak theory; gauge quotient; Higgs fixation; electromagnetic stabilizer; electric charge; Goldstone equivalence; custodial symmetry; electroweak normal form; AASC.

Proof and certification status

The manuscript separates scientific proof from certification. Imported interfaces retain their source authority. Stabilizer, response, neutral-basis, coupling, and charge results are proved locally. Standard QFT theorems concerning local gauge order parameters, BRST physical states, pole gauge independence, and longitudinal–Goldstone amplitudes are cited and specialized at an explicit target. The accompanying certification layer checks scope, dependency, provenance, encoded algebra, artifact identity, reproducibility, and the declared finite rival audit. It supports the theorem but does not substitute for a semantic proof of candidate completeness or exhaustion.

Exact nonclaim boundary

The result does not derive numerical electroweak parameters; deny the utility of gauge-fixed fields; identify every elementary-field correlator with a local gauge-invariant composite correlator nonperturbatively; settle the faithful global electromagnetic subgroup; construct the complete infrared charged Hilbert space; classify multi-Higgs or strongly coupled electroweak sectors; or prove the complete Standard Model symmetry atlas. Those are different endpoint types or successor targets.

Read this first: arc position and proof spine

SYM-8 is the vacuum and residual-symmetry stage of a dependency-ordered Standard Model symmetry program. SYM-0 fixes the typed status calculus; SYM-1 separates gauge architecture from quotient-trivial local operations and supplies the boundary exception grammar; SYM-5 separates physical flavor limits from weak-basis skin and fixed-tensor stabilizers; SYM-6 supplies the custodial packet and its deformation ancestry. The Higgs Structural Endpoint Arc fixes the bridge/fixation carrier $C_H = H_{\text{br}/\text{fix}}$, Higgs identity as admissible occupation of that carrier, and the nonpromotion of the field, vev, potential, pole, scale, or vacuum route into a second primitive identity. The present paper does not reopen those conclusions. It asks what the selected one-doublet fixation branch forces for the connected residual generator, gauge-response kernel, physical charge action, perturbative spectrum, Goldstone translation, and presentation-covariant electroweak normal form.

The arc-wide ordering rule is preserved:

framework-neutral QFT ledger $\longrightarrow$ quotient and observable certificates
 $\longrightarrow$ AASC structural role $\longrightarrow$ scope-preserving rival analysis.

A role label never supplies the QFT fact it later classifies. In particular, the words “skin,” “carrier,” and “standing” do not prove that a gauge operation is redundant, that a stabilizer is unique, that a pole is gauge independent, or that a longitudinal amplitude is represented by a Goldstone amplitude. Those burdens are discharged before the structural translation.

Scientific proof spine

1. Freeze the reference-complete target D_{EW}^{1H} , including field content, hypercharge convention, operator order, nonzero fixation branch, gauge quotient, boundary locus, perturbative regime, and the explicit spectral premise bundle $\mathcal{S}_{\text{pert}}$.
2. Establish that proper local gauge-related Higgs representatives lie in one physical fixation class and that no gauge-variant local representative is an ordinary physical order parameter for local gauge redundancy.
3. Compute the annihilator of the fixation representative and prove that its connected conjugacy class is one-dimensional, generated by $Q = T_3 + Y/(2y_H)$.
4. Construct the coordinate-free response form and coupling map from the Higgs kinetic term, prove $\ker \mu = C_g^{-1}(\mathbb{R}Q) = \mathbb{R}Q_g$ and $\text{rank } \mu = 3$, and diagonalize the neutral block into A_μ and Z_μ .
5. Reconstruct the charge-ratio ledger and the electromagnetic coupling projection, with normalization covariance explicitly proved.
6. Split residual electromagnetism into proper-local and boundary/asymptotic loci; attach a physical-action tag only where a nonzero quotient-level charge certificate exists.
7. Separate algebraic response from dynamical spectral existence, then reconstruct the certificate-conditional perturbative physical spectrum from $\mathcal{S}_{\text{pert}}$ and prove gauge-choice covariance of that packet.
8. State the longitudinal–Goldstone equivalence theorem with its energy regime, finite factor, and remainder; block double counting and all-energy extrapolation.
9. Import the Higgs carrier and custodial packet, then prove nonpromotion and electromagnetic/-custodial noncollapse.
10. Assemble the preceding results into the one-doublet electroweak normal-form theorem and audit a frozen finite rival registry without treating that registry as a proof of semantic completeness.

Claim hierarchy

The paper distinguishes four levels.

1. **Textbook-coordinate facts:** formulas in a chosen Higgs orientation, weak basis, neutral basis, and gauge fixing.
2. **Redescription-stable QFT facts:** fixation orbit, stabilizer conjugacy class, response rank and kernel, charge ratios, boundary-locus status, physical pole/cohomology packet, and amplitude relations.
3. **Structural role facts:** which entries are carrier constraints, physical actions, witnesses, representative skin, effective routes, obstructions, or target updates.
4. **Audit facts:** which declared registry rows, dependencies, encoded identities, and artifacts have been checked. These are integrity and coverage statements about the published project, not independent proofs that every scientifically possible rival has been generated.

A conclusion at one level is never promoted to another without a stated theorem or certificate.

Dependency-inversion exclusion

The local electroweak derivation is not permitted to assume the endpoint by naming the familiar Standard Model. Conversely, the endpoint is not required to rederive the entire Standard Model from no input. The target imports a fixed anomaly-neutral representation and charge architecture. SYM-8 proves what follows for the vacuum, residual action, response, and spectrum at that target. A rival that changes representation content, scalar content, global form, boundary conditions, operator category, or quantum regime is a target update, not a same-target counterexample.

One-sentence falsifier

The integrated result fails if a construction at exactly D_{EW}^{1H} satisfies the stated premises yet yields a different quotient fixation class, connected stabilizer, response rank or coupling-pullback kernel, charge-ratio ledger, residual-action locus, admitted physical spectrum, Higgs-carrier status, custodial status, or phase signature without breaking a cited theorem. An omitted same-target rival remains a legitimate challenge; the finite registry is not used to dismiss it.

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I

Scope, dependencies, and fixed target

1 Arc position, dependency order, and proof ownership

1.1 Why SYM-8 is a separate theorem burden

The local electroweak Lie algebra, the anomaly-neutral matter representation pattern, and the existence of a Higgs bridge/fixation carrier do not by themselves identify physical electromagnetism. A pre-vacuum realization may contain an Abelian charge coordinate without having selected the unique connected stabilizer of a nonzero one-doublet fixation branch. Likewise, a residual-sector syntax can record that a one-dimensional direction is available while deliberately stopping before it proves the generator Q , the photon direction, the weak mixing relation, or the physical charge action. This is the exact gap closed here.

The distinction matters because three different shortcuts would make the conclusion circular.

1. Writing $Q = T_3 + Y$ into the target as a definition would make the stabilizer theorem a restatement.
2. Importing the photon and Z as named mass eigenstates would make the response-kernel theorem a change of notation.
3. Calling the Higgs expectation value an order parameter for “broken gauge symmetry” would assume that a gauge-fixed representative carries ordinary physical multiplicity.

SYM-8 instead imports only the pre-vacuum representation, coupling, quotient, carrier, and residual-normal-form interfaces. It then computes the annihilator, response form, charge projection, spectrum packet, and lawful quotient locally.

1.2 Upstream interfaces

Table 1 records the dependencies actually used. The distinction between imported and locally proved content is part of the theorem statement, not merely a citation convention.

Table 1: Upstream interfaces and retained boundaries.

Source	Imported object	Use in SYM-8	Boundary retained
SYM-0 [1]	Typed symmetry record, quotient-action trichotomy, role vocabulary, target-update rule	Fixes the status fields and prohibits assigning a role before physical quotient and locus are known	Supplies no electroweak stabilizer or spectrum fact
SYM-1 [2]	Gauge architecture/operation split; proper-bulk quotient; boundary and large-operation separation; gauge-fixing no-authority	Governs local gauge quotient and residual electromagnetic locus split	Faithful global form and defect sectors remain open

Source	Imported object	Use in SYM-8	Boundary re-tained
SYM-3 [3]	Gauge-anomaly admissibility and obstruction transport	Keeps the target anomaly-neutral and prevents an anomalous candidate from being treated as a broken phase	Does not select the vacuum stabilizer
SYM-5 [4]	Weak-basis quotient; fixed-tensor stabilizer; invariant flavor data	Prevents Yukawa bases and rephasings from selecting the vacuum, charge ledger, or spectrum	Neutrino target remains minimal unless explicitly updated
SYM-6 [5]	Four custodial targets; bidoublet repackaging; diagonal $SU(2)_V$; deformation ancestry	Supplies the custodial interface and the noncollapse comparison with electromagnetism	Custodial exactness is target/limit dependent
SYM-7 [6]	Target discipline for accidental charges and operator order	Provides a consistency check for residual physical charge versus accidental global charges	No new SYM-8 theorem is imported
Higgs endpoint arc [7, 8, 9, 10, 11, 12]	$C_H = H_{br/fix}$; identity as carrier occupation; representative and route nonpromotion	Keeps the electroweak fixation and mass response carrier-indexed	Does not compute Q , A_μ , or the physical charge ledger
Gauge-vacuum compatibility [13]	Vacuum-supported response reduction and gauge-fixing exclusion	Allows a fixation representative while denying it independent selector authority	Does not identify physical electromagnetism
Residual-sector normal form [14]	Typed residual ledger after admissible vacuum-mediated reduction	Supplies pre-SYM-8 residual syntax and failure classes	Explicitly stops before Q , photon, weak mixing, and spectrum
Abelian charge-ratio rigidity [15]	Normalization-skin discipline and physical ratio standing	Governs hypercharge rescaling and final unit fixing	Does not choose the electroweak stabilizer
Standalone Standard Model [16]	Gauge product, representations, hypercharges, one-doublet bridge, anomaly/vacuum compatibility	Supplies the fixed representation and charge input used in the local calculation	Numerical parameters remain inputs
Realized Physical Interior [17]	Physical quotient, particle/pole role, vacuum and mass-response distinctions	Fixes what counts as physical spectral content	Does not prove all nonperturbative field/composite correspondences

Imported interface

The direct SYM dependency set is $\{\text{SYM0}, \text{SYM1}, \text{SYM5}, \text{SYM6}\}$. SYM-3 is imported through anomaly neutrality, and SYM-7 is used only as a consistency boundary. The Higgs and residual-

support manuscripts supply additional structural interfaces. None of these inputs contains the integrated SYM-8 electroweak normal-form theorem.

1.3 Mainstream quantum-field-theory inputs

The framework-neutral layer relies on standard results whose use is sharply limited.

- Elitzur’s theorem blocks an ordinary nonzero expectation value of a local gauge-variant field as a gauge-invariant order parameter without gauge fixing or an explicitly nonlocal construction [18]. It does not make gauge-invariant phase structure, Higgs response, or residual charge meaningless.
- BRST cohomology supplies a representation of the physical state space after gauge fixing in perturbative gauge theory [19, 20, 21]. It does not by itself prove the full interacting spectrum.
- Nielsen identities and their Standard Model implementations control gauge-parameter dependence and support gauge independence of physical pole data, including complex poles under the stated assumptions [22, 23].
- The longitudinal–Goldstone equivalence theorem relates high-energy amplitudes, with finite normalization factors and controlled corrections, rather than identifying fields or particles at all energies [24, 25, 26].
- Gauge-invariant composite-operator expansions provide a nonperturbative and perturbative bridge between gauge-invariant channels and familiar elementary-field poles in suitable regimes [27, 28]. This paper uses the bridge as an example and does not promote it to an unrestricted universal equality.
- Fradkin–Shenker continuity warns that gauge-fixed labels alone do not define a universal phase boundary in gauge–Higgs systems [29]. The present phase rule is therefore certificate-relative and does not claim a complete lattice phase diagram.

1.4 Division of evidentiary labor

The paper uses a three-column proof ledger.

1. **Physical antecedent:** transformation law, quotient, response form, current or charge, anomaly, pole, amplitude, or invariant operator.
2. **Structural translation:** anchor, carrier constraint, bridge, physical action, witness, skin, effective route, obstruction, or target update.
3. **Rival disposition:** lawful redescription, incompatible claim, incomplete certificate, target update, or open same-target burden.

A statement is counted as locally proved only when the first column is established in this paper. Imported theorems retain their source status. The integrated normal-form theorem composes the physical and structural results without changing their epistemic provenance. The finite rival audit records the status of declared examples but is not used as a candidate-completeness theorem.

1.5 Arc handoffs

SYM-8 fixes the connected local one-doublet vacuum, residual-action, response, and certificate-conditional spectrum normal form. It exports to SYM-9 the generator line $\mathbb{R}Q$, point-particle charge pairing, local response kernel, and the explicit statement that the faithful global electromagnetic subgroup, central intersections, line operators, monopoles, theta periodicities, and generalized symmetries remain unresolved. It exports to SYM-11 a finite set of typed atlas rows, a theorem-backed rival ledger, and the sector normal-form theorem. Candidate completeness and whole-Standard-Model exhaustion remain SYM-11 burdens.

Exact nonclaim boundary

No dependency in [table 1](#) is treated as evidence that the Standard Model is unique under arbitrary premises. The target is frozen before the local proof. The paper neither derives the observed representation pattern from the stabilizer nor derives numerical couplings from the charge generator.

2 Fixed target, target fibers, and convention lock**2.1 Reference-complete core target**

A vacuum or residual-symmetry claim is not well formed until the theory, quotient, phase, boundary, operator category, parameter stratum, and spectral premises are fixed. We therefore work with the target

$$D_{\text{EW}}^{1H} = (M^{1,3}, \mathfrak{g}_{\text{EW}}, \mathcal{F}_{\text{min}}, H, \mathcal{O}_{d \leq 4}, \mathcal{Q}_{\text{proper}}, \mathcal{B}, \mathcal{V}_{v \neq 0}, \mathcal{R}_{\text{pert}}, \mathcal{S}_{\text{pert}}, \mathcal{E}), \quad (1)$$

with the following fields.

Definition 2.1 (One-doublet electroweak target). The target D_{EW}^{1H} consists of:

1. fixed Minkowski background $M^{1,3}$;
2. local gauge algebra $\mathfrak{g}_{\text{EW}} = \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y$, while the faithful global quotient is recorded but deferred;
3. the minimal Standard Model chiral matter multiplets with no right-handed neutrino in the core target;
4. one complex scalar doublet $H \sim (\mathbf{1}, \mathbf{2})_{1/2}$ in the internal hypercharge convention;
5. the complete renormalizable operator grammar $\mathcal{O}_{d \leq 4}$, anomaly-neutral in the inherited representation;
6. the proper local gauge quotient $\mathcal{Q}_{\text{proper}}$, separating compact-support bulk operations from large, topological, or boundary-supported operations;
7. boundary and falloff data $\mathcal{B}$ sufficient to define the proper and asymptotic loci;
8. a selected nonzero fixation branch $\mathcal{V}_{v \neq 0}$, represented when convenient by $H_0 = (0, v)^T / \sqrt{2}$ with $v \neq 0$;
9. a weakly coupled perturbative regime $\mathcal{R}_{\text{pert}}$ in which BRST, pole, invariant-correlator, and high-energy amplitude certificates are meaningful;
10. the perturbative spectral certificate bundle $\mathcal{S}_{\text{pert}}$ of [theorem 2.2](#);
11. an evidence packet $\mathcal{E}$ recording all imported and local certificates.

The couplings g_2 and g_1 are nonzero but numerically unspecified.

Definition 2.2 (Perturbative spectral certificate bundle). The target datum $\mathcal{S}_{\text{pert}}$ comprises the dynamical premises needed to turn the algebraic response directions into physical perturbative sectors:

1. a nilpotent BRST complex and an admitted physical-state cohomology;
2. an admitted massless transverse electromagnetic field-strength pole sector;
3. admitted real or complex pole classes for the charged and neutral massive-vector channels in physical amplitudes;
4. an admitted gauge-invariant neutral scalar correlator/pole channel together with a nondegenerate radial Hessian on the selected branch;
5. the Nielsen-identity, gauge-parameter-transport, and weak-coupling hypotheses required for the stated pole and correlator comparisons.

These are target certificates, not consequences of the response-rank calculation. The local algebra fixes how the admitted sectors are organized, charged, and transported; it does not manufacture pole existence from rank alone.

The target contains a selected branch rather than a claim that a local gauge-variant expectation value is itself a physical observable. The representative H_0 is a coordinate used to compute the conjugacy class of the stabilizer and the response form. The physical fixation datum is its quotient class and the gauge-invariant response information it supports. Likewise, $\mathcal{S}_{\text{pert}}$ records the perturbative dynamical sector in which the response-to-spectrum bridge is licensed; it is not smuggled in through the endpoint theorem.

2.2 Controlled target fibers

Nearby questions are registered as distinct fibers, not silently mixed with the core target.

Table 2: Controlled target fibers.

Fiber	Change from D_{EW}^{1H}	Status in this paper
$D_{8,\text{bulk}}$	Proper compact-support local operations, fixed boundary	Core quotient locus
$D_{8,\partial}$	Nonvanishing admissible boundary/asymptotic parameter with differentiable charge	Mixed/physical action locus when certified
$D_{8,g'=0}$	Set $g_1 = 0$ in the appropriate custodial target	Custodial comparison fiber, not the physical core
D_{8,v_D}	Add right-handed neutrinos with Dirac structure	Controlled target update imported through SYM-6
$D_{8,v=0}$	Symmetric comparison point/branch	Comparison target; the nonzero-branch endpoint does not transport automatically
$D_{8,\text{ext}H}$	Add scalar multiplets, composite realization, or portal sector	Target update; requires new stabilizer and response analysis
$D_{8,\text{global}}$	Fix faithful global quotient, line and monopole spectra, defect category	SYM-9 burden
$D_{8,\text{nonpert}}$	Demand complete nonperturbative charged-state and spectrum construction	Open neighboring target, not claimed here

Proposition 2.3 (Target-update separation; SYM8-F01). *A change of scalar content, faithful global form, boundary class, operator category, neutrino field content, quantum regime, spectral certificate bundle, or fixation branch is not an alternative presentation of D_{EW}^{1H} . It defines a target fiber or a new target and cannot refute a same-target theorem without an explicit transport map preserving all target fields.*

Proof. Each listed datum occurs in the domain of at least one defining map: the physical quotient, representation action, stabilizer, response form, charge pairing, boundary generator, BRST complex, pole certificate, or candidate generator. Changing the datum changes the domain or codomain of that map, or changes the predicate by which a certificate is accepted. Hence the resulting object is not extensionally comparable under the same $\sim_{\text{adm}}$ relation. A comparison may still be scientifically relevant, but it is a target transport problem rather than a same-target rival. $\square$

2.3 Hypercharge normalization

Let the Abelian generator act on the Higgs as

$$YH = y_H H. \quad (2)$$

The internal convention is $y_H = \frac{1}{2}$. A doubled convention $\hat{Y} = 2Y$ has $\hat{y}_H = 1$. The covariant derivative is

$$D_\mu = \partial_\mu - ig_2 T^a W_\mu^a - ig_1 Y B_\mu. \quad (3)$$

Lemma 2.4 (Normalization covariance; SYM8-F02). *Under the simultaneous positive rescaling*

$$Y \mapsto Y' = \lambda Y, \quad g_1 \mapsto g'_1 = \frac{g_1}{\lambda}, \quad \lambda > 0,$$

the Abelian interaction $g_1 Y B_\mu$, the stabilizer line, the electric-charge ratios, and the response form are unchanged. The displayed generator becomes

$$Q = T_3 + \frac{1}{2y_H} Y = T_3 + \frac{1}{2y'_H} Y'.$$

A negative normalization change is the composition of this positive rescaling with the common generator/coupling sign convention and likewise changes no vertex or ratio.

Proof. The interaction pairing satisfies $g'_1 Y' = g_1 Y$. Since $y'_H = \lambda y_H$, the combination $Y'/(2y'_H)$ equals $Y/(2y_H)$. Every charge ratio is an eigenvalue of this invariant combination after the common electric unit is fixed. The response form depends on the representation-valued coupling map, which is likewise unchanged. Therefore the rescaling is a coordinate change in Abelian normalization, not a second physical charge structure. A simultaneous sign reversal preserves the product $g_1 Y$ and is separately recorded as a sign convention. $\square$

Convention firewall

Throughout the main text, $YH = \frac{1}{2}H$ and therefore $Q = T_3 + Y$. The formula $Q = T_3 + \hat{Y}/2$ is used only when $\hat{Y} = 2Y$ is explicitly declared. Both formulas are never attached to the same unqualified symbol.

2.4 Canonical field and generator notation

The $SU(2)$ generators in the doublet representation are $T^a = \sigma^a/2$, with

$$T^\pm = T^1 \pm iT^2, \quad W_\mu^\pm = \frac{W_\mu^1 \mp iW_\mu^2}{\sqrt{2}}.$$

For $g_1, g_2 > 0$, define

$$s_W = \frac{g_1}{\sqrt{g_1^2 + g_2^2}}, \quad c_W = \frac{g_2}{\sqrt{g_1^2 + g_2^2}}, \quad e = g_2 s_W = g_1 c_W, \quad (4)$$

and

$$A_\mu = s_W W_\mu^3 + c_W B_\mu, \quad Z_\mu = c_W W_\mu^3 - s_W B_\mu. \quad (5)$$

The symbols s_W, c_W are structural coordinates determined by the nonzero couplings. No numerical value of the weak angle is claimed.

2.5 Exact claim ceiling

The target supports algebraic masses

$$m_W^2 = \frac{g_2^2 v^2}{4}, \quad m_Z^2 = \frac{(g_1^2 + g_2^2) v^2}{4}, \quad m_A^2 = 0,$$

as consequences of the fixed representative and response form. They are quadratic-response formulas in the selected target, not by themselves proofs of isolated pole existence and not numerical predictions of v , g_1 , g_2 , pole masses, widths, or radiative corrections. Pole existence and gauge-independent identification are supplied only when the corresponding entries of S_{pert} are present.

Exact nonclaim boundary

The target does not choose the exact global group $[SU(2)_L \times U(1)_Y]/\Gamma$, does not specify line operators, and does not infer an exact global electromagnetic subgroup from the Lie algebra alone. The connected local stabilizer theorem proved below is compatible with multiple faithful global completions; those are separated in SYM-9.

3 Lawful presentations, rival loci, and the electroweak normal form

3.1 Why equality of formulas is the wrong equivalence relation

The same physical electroweak structure can be written in unitary gauge or an R_ξ gauge, in different weak-flavor bases, with different hypercharge normalizations, with (W^3, B) or (A, Z) neutral coordinates, or with the Higgs doublet repackaged as a custodial bidoublet. Literal equality would count those displays as different. An unrestricted assertion of “the same physics” would be too weak because it could conceal a changed boundary action, anomaly status, current pairing, spectral premise, response kernel, or carrier. The comparison relation must therefore transport every dependent object together and preserve the physical quotient and target fingerprint.

Definition 3.1 (Lawful electroweak presentation change). A presentation change at D_{EW}^{1H} is lawful when it is invertible on its declared presentation domain and preserves the complete physical and structural interfaces used in this paper. The admitted families are:

1. proper gauge transformations preserving the target and boundary data, acting simultaneously on the Higgs representative, annihilator, response form, and phase representative;
2. weak-flavor basis changes with the Yukawa and invariant flavor data transported as in SYM-5;
3. hypercharge normalization $Y \mapsto \lambda Y$ with contragredient coupling transport $g_1 \mapsto g_1/\lambda$, together with the corresponding generator and charge-coordinate change;
4. kinetic-metric-aware neutral-field basis changes, with the response form, current pairing, and spectrum labels transported by the same basis map;
5. gauge-fixing and Goldstone-chart changes that preserve BRST cohomology, physical amplitudes, invariant correlators, admitted pole classes, and the selected phase signature;
6. the invertible one-doublet repackaging $H \leftrightarrow \Phi = (\tilde{H}, H)$, with the Higgs carrier and custodial target left unchanged.

A transformation of only one displayed component is not lawful when its dependent interfaces are left behind.

Definition 3.2 (Admissible presentation equivalence). Two electroweak presentations are related by $\sim_{\text{adm}}$ when they are connected by a finite composition of lawful presentation changes from [theorem 3.1](#). This relation identifies representations of one physical construction; it does not identify target changes, failed physical certificates, or genuinely different quotient-level content.

3.2 Canonical normal-form packet

The endpoint is not one preferred gauge or one preferred Lagrangian display. It is represented by the packet

$$\mathfrak{E}_{\text{EW}} = \left([H_0]_{\mathcal{G}_{0,\partial}}, [u(1)_Q]_{\text{conj}}, [Q]_{\text{norm}}, C_g, [\mu_{H_0}]_K, \mathcal{Q}_{\text{phys}}, \mathcal{A}_{\text{em}}, \mathcal{P}_{\text{phys}}, \mathcal{T}_{G/L}, C_H, C_{\text{cust}}, \Sigma_{\text{phase}} \right). \quad (6)$$

The entries are the proper-gauge fixation orbit; connected stabilizer conjugacy class; normalized electromagnetic generator class; coupling map; response form modulo kinetic-metric-aware basis transport; physical charge-ratio ledger; locus-split electromagnetic action; certificate-conditional perturbative spectrum; Goldstone–longitudinal translation; Higgs bridge/fixation carrier; custodial target and deformation packet; and invariant phase signature. The evidence, dependency, provenance, and reproducibility records certify the construction but are not themselves components of the physical packet.

Lemma 3.3 (Presentation covariance). *Every lawful presentation change preserves the target, the physical quotient, and the invariant entries of $\mathfrak{E}_{\text{EW}}$. Consequently, if $E \sim_{\text{adm}} E'$, then E and E' represent the same electroweak normal form.*

Proof. Proper gauge transformations preserve the quotient fixation class and conjugate the annihilator and response data together. Weak-basis changes preserve flavor invariants and do not alter Q . Hypercharge normalization preserves $g_1 Y$, the annihilator line, and charge ratios. Neutral-basis changes act by the corresponding congruence and dual current transformation and therefore preserve rank, kernel, and interaction pairing. Gauge-fixing changes preserve the physical cohomology, amplitudes, invariant correlators, and admitted poles under the stated QFT hypotheses. The bidoublet map is an invertible repackaging of the same four real Higgs components and preserves C_H . Each generator of $\sim_{\text{adm}}$ therefore preserves the listed invariant entries, and so does every finite composition. $\square$

3.3 Rival loci

A serious rival must first identify the physical locus in which it differs. The finite audit supplied with this paper is organized by the following ten loci.

Protocol 3.4 (Rival-generation and disposition protocol). For each locus in table 3, a proposed rival must state its target, acting object, acted-upon object, quotient status, anomaly status, vacuum status, boundary or global-form dependence, response or spectrum data, and claimed structural role. It is then classified as:

1. a lawful redescription under $\sim_{\text{adm}}$;
2. incompatible with a proved or imported physical interface;
3. certificate-incomplete;
4. a controlled target fiber or explicit target update; or
5. an open same-target burden.

A proposed same-target rival is never rejected merely because it is absent from the published registry. Its absence is a coverage challenge, not a physical refutation.

3.4 Certification boundary

The project includes a manually curated fifty-row rival registry. Each row records the target relation, decisive theorem or certificate, and audit disposition for a concrete challenge. This is useful evidence that the principal failure modes have been addressed consistently and that the manuscript, tables, and source ledger agree. It is not a proof that the ten loci contain every semantically possible electroweak construction, nor that a finite table establishes universal uniqueness. Those stronger claims require an independent candidate-completeness theorem and belong to the SYM-11 capstone.

Table 3: Rival loci used by the SYM-8 audit.

Code	Rival locus	Typical challenge
VAC	Vacuum/fixation	a second vacuum from a gauge-related Higgs orientation; a vev as primitive selector
STAB	Stabilizer	a different connected generator; a representative-dependent subgroup
RESP	Response form	a second null direction; a different rank; basis-dependent mass claim
CHG	Charge	a different charge ratio; normalization promoted to physics
ACT	Action locus	a local residual operation called physical without boundary charge; a boundary action called pure gauge
SPEC	Spectrum	a gauge-dependent pole or field component treated as an independent particle
GOLD	Goldstone	an additional physical-particle count; an exact all-energy field identity
HIGGS	Higgs role	the field, vev, potential, or pole promoted over the bridge/-fixation carrier
CUST	Custodial	$SU(2)_V$ identified with $U(1)_{em}$; a diagnostic promoted to a definition
PHASE	Phase/chart	a gauge-fixing chart counted as a distinct phase; an invariant phase change dismissed as notation

Certificate doctrine

The scientific normal-form theorem is established by the mathematical and physical arguments in the main text. The finite registry, validators, hashes, and hostile checks certify scope, dependency, provenance, encoded algebra, artifact identity, reproducibility, and the declared audit. They support the theorem but do not substitute for a semantic proof of completeness or exhaustion.

II

Framework-neutral electroweak reconstruction

4 Gauge quotient and the local-gauge non-order-parameter theorem

4.1 The language problem

A gauge-fixed electroweak calculation usually begins by choosing an orientation

$$H_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}, \quad v > 0.$$

The choice is efficient: it makes the vector response matrix transparent, separates a radial coordinate from angular coordinates, and permits familiar perturbative gauges. The efficiency does not settle the physical status of the coordinate. A local gauge operation changes H_0 while preserving the physical gauge orbit, and an unfixed local gauge-variant one-point function is not an ordinary gauge-invariant

order parameter. The correct object is therefore a fixation class with gauge-invariant response certificates, not an isolated vector in field-coordinate space.

4.2 Proper local gauge quotient

Let $\mathcal{X}_{v \neq 0}$ denote the regular nonzero-Higgs configurations in the selected perturbative neighborhood, and let $\mathcal{G}_{0,\partial}$ be the group of smooth electroweak gauge transformations that are homotopic to the identity in the admitted bulk locus and equal to the identity at the boundary specified by $\mathcal{B}$. The quotient map is

$$q_{\text{proper}} : \mathcal{X}_{v \neq 0} \longrightarrow \mathcal{X}_{v \neq 0} / \mathcal{G}_{0,\partial}.$$

A transformation outside $\mathcal{G}_{0,\partial}$ is not assigned this status automatically. Large, topological, or boundary-supported transformations are separate records.

Definition 4.1 (Fixation class). A *fixation class* is a quotient class $[H]_{\mathcal{G}_{0,\partial}}$ together with the gauge-invariant local response and spectrum certificates required by D_{EW}^{1H} . A bare expectation value in a chosen gauge is a representative coordinate of the class, not the whole fixation datum.

Theorem 4.2 (Proper local gauge quotient). *For every $g \in \mathcal{G}_{0,\partial}$ and every admitted configuration $x \in \mathcal{X}_{v \neq 0}$,*

$$q_{\text{proper}}(g \cdot x) = q_{\text{proper}}(x).$$

Consequently, two Higgs orientations related by a proper local gauge transformation do not define two physical electroweak vacua at the core locus.

Proof. This is the defining quotient relation, but its applicability is nontrivial: anomaly neutrality, fixed boundary data, and the exclusion of large/topological transformations are required for the operation to be an admitted proper gauge arrow. Those conditions are part of D_{EW}^{1H} and the imported SYM-1 interface. Once they hold, gauge-related representatives lie in the same orbit by construction, and every physical observable admitted by the quotient has the same value on the two representatives. $\square$

4.3 Order parameters and gauge variance

An ordinary order parameter for spontaneous breaking of a global symmetry is a physical observable or a limit of observables whose vacuum expectation value transforms nontrivially under a physical action. A local gauge-variant field component fails that criterion at the proper bulk locus because the local gauge operation is quotient-trivial. Elitzur's theorem supplies the standard quantum-field-theoretic obstruction to interpreting an unfixed local gauge-variant expectation value as a nonzero gauge-invariant witness [18].

This does not prohibit all vacuum diagnostics. Gauge-invariant composites, response ranks, pole spectra, long-range behavior, Wilson or line observables when admitted, and boundary charges can distinguish physical regimes. Nor does it prohibit gauge fixing. It prohibits the inference

chosen nonzero gauge-fixed coordinate

$$\not\Rightarrow \begin{array}{l} \text{ordinary physical breaking} \\ \text{of local gauge redundancy.} \end{array}$$

Theorem 4.3 (Local-gauge non-order-parameter theorem; SYM8-Q01). *At $D_{8,\text{bulk}}$, no local gauge-variant Higgs representative $H(x)$, gauge-fixed one-point coordinate $\langle H \rangle_\xi$, or gauge orientation can serve by itself as an ordinary physical spontaneous-breaking order parameter for local electroweak gauge redundancy. Its admissible physical use factors through the proper-gauge fixation class and gauge-invariant response, spectrum, or correlation certificates.*

Proof. Suppose a submitted local gauge-variant quantity O were an ordinary physical order parameter for the local gauge action. Then there would exist $g \in \mathcal{G}_{0,\partial}$ and a physical quotient class $[x]$ such that the

transformed value distinguished $[g \cdot x]$ from $[x]$. But [theorem 4.2](#) gives $[g \cdot x] = [x]$. Thus the difference is not a difference on the physical quotient. In a gauge-fixed representation, a nonzero value can still be used as a coordinate, but its physical conclusions must be transported to quotient-stable response or observable data. The local gauge-variant quantity alone therefore fails the physical-order-parameter criterion. $\square$

Corollary 4.4 (Gauge-fixing no-authority). *A gauge-fixing functional, gauge parameter ξ , or unitary-gauge elimination of angular coordinates cannot select the physical residual generator, create an additional vacuum, change the response rank, or alter the charge-ratio ledger without violating the physical certificates of D_{EW}^{1H} .*

Proof. Gauge fixing chooses a section or local coordinate system on the representative space. A lawful change of gauge lies in $\text{Pres}_8(D_{EW}^{1H})$ only when the physical cohomology, pole, S-matrix, and invariant-correlation certificates are preserved. Hence it cannot alter the listed quotient-level data. $\square$

4.4 Vacuum orbit versus vacuum manifold shorthand

The phrase “vacuum manifold” may refer to several objects: minima of a gauge-fixed potential, the orbit of a representative under the gauge group, the quotient of minima by proper gauge operations, or a global-symmetry coset supporting physical Goldstone modes. These objects need not coincide. In the core one-doublet gauge problem, the orbit of H_0 is useful for computing the stabilizer, but points on the proper-gauge orbit are not distinct physical vacua. A genuine global symmetry may still be spontaneously realized in a controlled target; the custodial interface in [section 12](#) is treated separately.

Standard phrase and preserved meaning		
Standard phrase	Preserved meaning	Blocked promotion
“Choose the Higgs vacuum”	Choose a convenient representative of the selected nonzero fixation class	Representative orientation as a second physical vacuum
“Break $SU(2)_L \times U(1)_Y$ ”	The Higgs-induced response form has a three-dimensional image and a one-dimensional stabilizer	Ordinary spontaneous breaking of quotient-trivial local operations
“Unitary gauge removes Goldstones”	A gauge choice eliminates angular coordinates from that presentation	Physical particles literally deleted by a representative choice
“Residual $U(1)$ ”	A connected annihilator and electromagnetic physical-action packet survive	Every local residual gauge operation becomes a physical global action

4.5 What remains physically nontrivial

The theorem removes multiplicity from gauge-related representatives; it does not remove the gauge architecture that licenses transport, curvature, charge, and the response form. The following objects remain nontrivial:

- the representation map of $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y$ on the matter and Higgs fields;
- the connection, curvature, and covariant derivative;
- anomaly cancellation and BRST consistency;
- the annihilator and response form induced by the nonzero fixation class;
- the global or boundary charge action where the quotient locus admits it;

- the gauge-invariant or gauge-independent physical spectrum.

The architecture/operation distinction is therefore positive: quotient-trivial operations coexist with load-bearing gauge structure.

4.6 Controlled Abelian-Higgs comparison

A one-complex-scalar Abelian model makes the quotient logic visible without the non-Abelian book-keeping. Let

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + |D_\mu\varphi|^2 - V(|\varphi|^2), \quad D_\mu\varphi = (\partial_\mu - igA_\mu)\varphi.$$

A gauge-fixed chart may write

$$\varphi(x) = \frac{1}{\sqrt{2}}(v + h(x))e^{i\pi(x)/v}.$$

The local phase of φ changes under $\varphi \mapsto e^{i\alpha(x)}\varphi$, while the gauge field transforms so that $D_\mu\varphi$ is covariant. For a proper local transformation the pair (φ, A_μ) moves within one gauge orbit. The representative value $\varphi = v/\sqrt{2}$ therefore cannot by itself distinguish physical vacua. Nevertheless, insertion into the kinetic term produces

$$|D_\mu\varphi|^2 \supset \frac{1}{2}g^2v^2A_\mu A^\mu,$$

which is a nonzero response in the sole gauge direction. The response algebra is one-dimensional and its kernel is zero. There is no residual connected local gauge direction in this toy target, even though a gauge-fixed nonzero scalar coordinate exists. This comparison isolates two lessons.

First, a nonzero representative does not automatically imply a residual group. The residual structure depends on the annihilator of the representation. Second, quotient-triviality of local gauge operations does not erase the response. Gauge-invariant observables such as $|\varphi|^2$, suitable field-strength correlations, and physical pole data can distinguish the selected regime while the local phase remains representative.

The electroweak calculation differs because the one-doublet representation has a one-dimensional annihilator. The same logic therefore yields a rank-three form rather than a full-rank one. The photon is not inferred from the word “Higgs”; it is the kernel forced by the electroweak weights.

4.7 Gauge-invariant witnesses and nonlocal dressings

A local gauge-variant expectation value is not the only possible vacuum witness. Depending on the target, one may use:

- expectation values or correlators of local gauge-invariant composites such as $H^\dagger H$;
- the spectrum and pole structure of gauge-invariant channels;
- the rank and kernel of the response form, expressed invariantly;
- Wilson, 't Hooft, or other extended operators when the operator category admits them;
- asymptotic charges and long-range fields at a specified boundary;
- nonlocal dressed charged fields with an explicitly declared dressing and infrared regime.

These witnesses answer different questions. $H^\dagger H$ can detect radial structure but does not by itself identify the electromagnetic generator. A Wilson loop can probe screening or confinement properties but depends on representation, global form, and line-operator content. A dressed charged field can create a gauge-invariant charged excitation but introduces a nonlocal or asymptotic structure that is not a local order parameter. The response form and charge ledger used in this paper are selected because they are exactly the certificates required for the one-doublet residual problem.

This distinction prevents a common overcorrection. From “a local gauge-variant vev is not a physical order parameter” it does not follow that the gauge-Higgs regime has no invariant physics. What follows is that the physical statement must be moved to the correct observable or quotient level. The entire SYM-8 construction is that move, performed without discarding the efficient representative language.

4.8 Ward identities and the architecture/operation split

Gauge architecture constrains the quantum theory even when proper local operations are quotient-trivial. Infinitesimal gauge redundancy gives identities among Green functions and, after gauge fixing, Slavnov–Taylor or BRST identities. Anomaly cancellation is required for those identities to define a consistent quantum gauge system. The Higgs-induced response form is built from the covariant derivative fixed by that architecture. The electromagnetic Ward identities and charge conservation in the admitted perturbative sectors are then associated with the residual generator and its physical action.

These facts explain why “gauge is only redundancy” is too compressed. The operation can be a redundancy while the differential identities, representation constraints, connection, curvature, charge pairing, and boundary generators are indispensable. The role map therefore assigns skin to the proper local operation, not to the complete gauge architecture.

4.9 Representative utility theorem

The no-order-parameter theorem is compatible with unrestricted use of a gauge-fixed representative when the following transport condition is met. Let s_ξ be a local gauge-fixing section labeled by ξ , and let F_ξ be a quantity computed in that chart. The quantity has admissible physical use when there exists a quotient-level object F_{phys} and a certificate

$$\tau_\xi : F_\xi \mapsto F_{\text{phys}}$$

such that for any two admitted charts ξ, ξ' ,

$$\tau_\xi(F_\xi) = \tau_{\xi'}(F_{\xi'}).$$

Examples include a gauge-independent complex pole, an S-matrix element, the response rank, or a gauge-invariant correlator. The chart quantity need not be invariant pointwise; it must have a verified descent. This is the precise sense in which gauge-fixed calculations are preserved while representative promotion is blocked.

Failure condition

Theorem 4.3 would fail at the declared locus if a proper local gauge operation changed a certified physical observable, boundary charge, topological sector, or physical state. Such an operation would then have been misclassified as proper-local and must be moved to the appropriate boundary, large, or topological record. The theorem is not protected by relabeling; it is protected only by the explicit locus conditions.

5 The connected electromagnetic stabilizer

5.1 Coordinate-free annihilator

Let $V_H \cong \mathbb{C}^2$ be the Higgs representation space and

$$\rho_H : \mathfrak{g}_{\text{EW}} \longrightarrow \mathfrak{u}(V_H)$$

the infinitesimal representation, including the hypercharge action. For a nonzero representative $h \in V_H$, define

$$\text{Ann}(h) = \{X \in \mathfrak{g}_{\text{EW}} \mid \rho_H(X)h = 0\}. \quad (7)$$

If $h' = g \cdot h$, equivariance gives

$$\text{Ann}(h') = \text{Ad}_g \text{Ann}(h).$$

Thus the literal subalgebra depends on orientation, but its conjugacy class does not. The physical residual datum is the connected conjugacy class, together with its action on the inherited representation and its boundary locus.

5.2 Explicit calculation

Choose

$$H_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}, \quad v \neq 0,$$

and write

$$X = a_1 T_1 + a_2 T_2 + a_3 T_3 + b Y.$$

Using $T_a = \sigma_a/2$ and $YH = y_H H$,

$$XH_0 = \frac{v}{2\sqrt{2}} \begin{pmatrix} a_1 - ia_2 \\ -a_3 + 2by_H \end{pmatrix}. \quad (8)$$

The annihilator equations are

$$a_1 = 0, \quad a_2 = 0, \quad a_3 = 2by_H.$$

Hence

$$\text{Ann}(H_0) = \mathbb{R} \left(T_3 + \frac{1}{2y_H} Y \right). \quad (9)$$

Theorem 5.1 (Connected stabilizer uniqueness; SYM8-Q02). *For the nonzero one-doublet fixation class in D_{EW}^{1H} , the connected annihilator in the electroweak gauge algebra is one-dimensional. Its conjugacy class is represented by*

$$\mathfrak{u}(1)_Q = \mathbb{R}Q, \quad Q = T_3 + \frac{1}{2y_H} Y.$$

In the convention $y_H = \frac{1}{2}$, $Q = T_3 + Y$. Any other connected one-dimensional stabilizer of the same fixation class and representation is conjugate to this line; any stabilizer of different dimension changes the target or the fixation stratum.

Proof. Equation (8) shows that the annihilator of the representative is exactly one-dimensional and is spanned by Q . For any other representative $H'_0 = gH_0$ in the same proper-gauge orbit, equivariance gives $\text{Ann}(H'_0) = \text{Ad}_g \text{Ann}(H_0)$. Thus the connected conjugacy class is independent of orientation. A submitted stabilizer with an additional independent generator would contain an $X \notin \mathbb{R}Q$ annihilating H_0 , contradicting (8). A zero-dimensional stabilizer would omit Q , which directly annihilates H_0 . Therefore the connected stabilizer class is unique at the fixed target. $\square$

Corollary 5.2 (Generator normalization). *The stabilizer theorem fixes a line, not an absolute unit. Multiplying Q by a nonzero scalar changes the charge unit and coupling coordinate inversely. The physical point-particle charge ratios are fixed after one conventional unit, such as $Q(e_R) = -1$, is chosen.*

Corollary 5.3 (Representative covariance). *The statement “the lower component is neutral” is coordinate-dependent, while the statement “the fixation class has a one-dimensional annihilator whose inherited action yields the electromagnetic charge ratios” is redescription-stable.*

5.3 Connected local result versus faithful global subgroup

The calculation takes place in the Lie algebra and determines the identity component locally. Exponentiating Q inside a chosen global gauge group may involve central identifications and a nontrivial intersection with the faithful matter representation. Different quotients of $SU(2)_L \times U(1)_Y$ can share the same local algebra and point-particle action while differing in line operators or monopole sectors. Accordingly, the theorem is stated as a connected local stabilizer theorem. The exact global subgroup and its defect-sensitive completion are exported to SYM-9.

Exact nonclaim boundary

Theorem 5.1 does not prove that the faithful residual group is literally a direct-product copy of $U(1)$ for every global-form target. It proves the one-dimensional connected annihilator and its action on the admitted local field representation. Global identifications remain target data.

5.4 Abelian-Higgs comparison

The same logical structure already appears in the Abelian Higgs model. A nonzero charged scalar representative can be fixed in a convenient gauge, but the gauge orientation is not a physical vacuum label. The response form can acquire full rank in the one-dimensional gauge algebra, leaving no connected local gauge stabilizer. The comparison shows that “nonzero scalar” does not automatically imply a residual $U(1)$; the electroweak residual direction is a consequence of the representation weights and the one-doublet annihilator equations.

5.5 Falsification test

A genuine same-target counterexample must supply a second line $\mathbb{R}X \subset \mathfrak{g}_{EW}$ that annihilates the same nonzero one-doublet fixation class, acts consistently on the inherited fields, and is not conjugate or normalization-equivalent to $\mathbb{R}Q$. Equation (8) makes the burden explicit: such an X must solve the three annihilator equations without being proportional to Q . No such solution exists.

6 Gauge-response rank, kernel, and the neutral A/Z normal form**6.1 Coordinate-free response form and coupling map**

The Higgs kinetic term

$$(D_\mu H)^\dagger D^\mu H$$

induces a positive-semidefinite quadratic form on the real gauge-field coefficient space. It is important not to identify that space literally with the symmetry-generator algebra after the couplings have been inserted. Let $\mathfrak{a}_{EW}$ denote the real coefficient space with basis represented by (W^1, W^2, W^3, B) , and define the coupling isomorphism

$$C_g : \mathfrak{a}_{EW} \longrightarrow \mathfrak{g}_{EW}, \quad C_g(w^a T_a + bY) = g_2 w^a T_a + g_1 bY. \quad (10)$$

Because $g_1, g_2 \neq 0$, C_g is invertible. Define

$$L_{H_0}(u) = \rho_H(C_g u) H_0, \quad u \in \mathfrak{a}_{EW},$$

and

$$\mu_{H_0}(u, w) = 2 \operatorname{Re} \langle L_{H_0}(u), L_{H_0}(w) \rangle. \quad (11)$$

This is the gauge-boson mass-response form in a coordinate-free guise. It is not a numerical pole-mass matrix; it is the tree-level quadratic response induced by the fixed branch. The map C_g is part of the response certificate because it links the field-coordinate kernel to the generator that acts on matter.

Lemma 6.1 (Kernel–stabilizer correspondence). *For nonzero g_1, g_2 ,*

$$\ker \mu_{H_0} = C_g^{-1}(\operatorname{Ann}(H_0)) = \mathbb{R}Q_g, \quad Q_g := C_g^{-1}(Q). \quad (12)$$

In the general hypercharge convention,

$$Q_g = \frac{1}{g_2} T_3 + \frac{1}{2y_H g_1} Y.$$

For $y_H = \frac{1}{2}$, its neutral coefficient vector is proportional to (g_1, g_2) , the photon null vector in the (W^3, B) basis.

Proof. Positive semidefiniteness gives $\mu_{H_0}(u, u) = 2\|\rho_H(C_g u)H_0\|^2$. Hence $u \in \ker \mu_{H_0}$ exactly when $C_g u \in \text{Ann}(H_0)$. This proves the first equality in (12). Since C_g is an isomorphism and $\text{Ann}(H_0) = \mathbb{R}Q$ by [theorem 5.1](#), its inverse image is the one-dimensional line $\mathbb{R}C_g^{-1}(Q) = \mathbb{R}Q_g$. Applying C_g^{-1} to the coefficients of $Q = T_3 + Y/(2y_H)$ gives the displayed formula. In the core convention, $(1/g_2, 1/g_1)$ is proportional to (g_1, g_2) , as claimed. $\square$

Theorem 6.2 (Gauge-response rank and kernel; SYM8-Q03). *At D_{EW}^{1H} , the Higgs-induced gauge-response form has*

$$\text{rank } \mu_{H_0} = 3, \quad \dim \ker \mu_{H_0} = 1.$$

The kernel is the coupling pullback $\mathbb{R}Q_g = C_g^{-1}(\mathbb{R}Q)$ of the electromagnetic stabilizer line, and the image contains two charged and one neutral massive response directions.

Proof. The gauge-field coefficient space has real dimension four. By [theorem 6.1](#), its kernel is one-dimensional. Rank-nullity therefore gives rank three. Because C_g is an isomorphism, the field-coordinate preimages of the two charged generator directions are outside the kernel. In the neutral coefficient plane, any line independent of $\mathbb{R}Q_g$ maps to a generator outside $\text{Ann}(H_0)$, giving the remaining neutral image direction. Thus the image decomposes into two charged and one neutral response directions. $\square$

6.2 Canonical matrix calculation

With $y_H = \frac{1}{2}$, insertion of H_0 into the kinetic term yields

$$\mathcal{L}_{\text{resp}} = \frac{g_2^2 v^2}{4} W_\mu^+ W^{-\mu} + \frac{v^2}{8} (g_2 W_\mu^3 - g_1 B_\mu)^2. \quad (13)$$

In the neutral basis (W_μ^3, B_μ) , the mass-squared matrix is

$$M_0^2 = \frac{v^2}{4} \begin{pmatrix} g_2^2 & -g_2 g_1 \\ -g_2 g_1 & g_1^2 \end{pmatrix}. \quad (14)$$

Its determinant vanishes and its trace is $v^2(g_1^2 + g_2^2)/4$. The normalized eigenvectors are

$$\begin{pmatrix} s_W \\ c_W \end{pmatrix} \quad \text{with eigenvalue } 0, \quad \begin{pmatrix} c_W \\ -s_W \end{pmatrix} \quad \text{with eigenvalue } \frac{v^2}{4}(g_1^2 + g_2^2).$$

They define A_μ and Z_μ as in (5).

Theorem 6.3 (Neutral normal form; SYM8-Q04). *Among neutral field coordinates with canonical kinetic metric, the response form has one and only one null direction. The normalized field-coordinate representative of Q_g is, up to sign and lawful orthogonal basis transport,*

$$A_\mu = s_W W_\mu^3 + c_W B_\mu,$$

and the orthogonal image direction is

$$Z_\mu = c_W W_\mu^3 - s_W B_\mu.$$

The associated tree-level response eigenvalues are

$$m_A^2 = 0, \quad m_Z^2 = \frac{v^2}{4}(g_1^2 + g_2^2),$$

while $m_W^2 = g_2^2 v^2/4$.

Proof. The explicit diagonalization gives the stated eigenvectors and eigenvalues. Uniqueness follows because the neutral matrix has rank one for nonzero v, g_1, g_2 ; its nullspace is therefore one-dimensional. Any kinetic-metric-preserving change of neutral coordinates carries the matrix by orthogonal congruence and transports its kernel and image without changing their dimensions. Thus a different lawful basis gives the same null line and massive line, not a second physical photon or Z . $\square$

6.3 Basis covariance and interaction pairing

Not every invertible 2×2 matrix is an admissible neutral-field redescription. A lawful arrow must transport the kinetic metric and the coupling-to-current map together. Let N be a neutral coordinate change. If the field vector transforms as $V' = NV$, then the kinetic form, response form, and current coupling must be pushed forward consistently. A candidate that diagonalizes the response matrix while changing the current pairing is not the same physical packet.

Proposition 6.4 (Full neutral-packet covariance). *Let (K, M, J) be the neutral kinetic metric, response matrix, and current-coupling covector. Under an invertible change $V' = NV$, the transported packet is*

$$K' = N^{-T}KN^{-1}, \quad M' = N^{-T}MN^{-1}, \quad J' = JN^{-1}.$$

If N preserves K and all three objects are transported, then kernel, image, pole content, and current pairing are equivalent. Testing only one eigenvector is insufficient.

Proof. Substitution of $V = N^{-1}V'$ into the quadratic kinetic and response terms and the linear current coupling gives the formulas. Congruence by an invertible matrix preserves rank and maps the kernel bijectively. The current pairing is preserved only when the covector is transported as shown. $\square$

Response certificate

The canonical response certificate records: matrix dimensions; symmetry and positive semidefiniteness; determinant and rank; an explicit kernel basis; a complementary image basis; the kinetic metric; the forward and inverse basis maps; and both transformed current pairings. The project validator checks these identities symbolically.

6.4 Degree-of-freedom identity

Before fixation, four massless gauge fields contribute $4 \times 2 = 8$ transverse polarizations, and one complex Higgs doublet contributes four real scalar coordinates, for a total of twelve perturbative coordinate degrees of freedom. After the response reorganization, one massless vector contributes two polarizations, three massive vectors contribute nine, and one radial scalar contributes one, again totaling twelve:

$$4 \cdot 2 + 4 = 1 \cdot 2 + 3 \cdot 3 + 1 = 12.$$

The identity is a bookkeeping check, not an independent proof of the physical Hilbert space. The BRST and spectrum certificates in [section 9](#) supply the physical interpretation.

Exact nonclaim boundary

The tree-level response eigenvalues are not asserted to be exact pole masses. Radiative corrections, renormalization schemes, unstable-particle complex poles, and nonperturbative effects belong to the spectrum certificate. The rank/kernel theorem is the structural content used by the endpoint.

7 Electromagnetic generator, coupling projection, and charge-ratio ledger

7.1 The generator and the neutral interaction

In the internal convention, $Q = T_3 + Y$. Inverting (5) gives

$$W_\mu^3 = s_W A_\mu + c_W Z_\mu, \quad B_\mu = c_W A_\mu - s_W Z_\mu.$$

The neutral interaction becomes

$$g_2 T_3 W_\mu^3 + g_1 Y B_\mu = (g_2 s_W T_3 + g_1 c_W Y) A_\mu + (g_2 c_W T_3 - g_1 s_W Y) Z_\mu \quad (15)$$

$$= e Q A_\mu + \frac{g_2^2}{c_W} (T_3 - s_W^2 Q) Z_\mu. \quad (16)$$

The first term links the massless field-coordinate kernel $\mathbb{R}Q_g$ to the generator line $\mathbb{R}Q$ through C_g , and shows that the normalized massless direction couples universally with e to the physical electric-charge operator.

Proposition 7.1 (Electromagnetic coupling projection). *For nonzero g_1, g_2 , the neutral interaction admits the unique kernel/image decomposition (16). The coefficient of the massless direction is eQ , where $e = g_2 s_W = g_1 c_W$. Any lawful neutral-basis or hypercharge-normalization change transports this pairing without changing the charge ratios.*

Proof. Substitute the inverse rotation and use $Q = T_3 + Y$, $g_2 s_W = g_1 c_W = e$, and $g_1 s_W = g_2 s_W^2 / c_W$. Uniqueness follows from the one-dimensional kernel and image of the neutral response matrix together with the fixed current-coupling map. $\square$

7.2 Matter and Higgs charge ratios

The inherited one-generation representation ledger is

$$Q_L \sim (\mathbf{3}, \mathbf{2})_{1/6}, \quad u_R \sim (\mathbf{3}, \mathbf{1})_{2/3}, \quad d_R \sim (\mathbf{3}, \mathbf{1})_{-1/3},$$

$$L_L \sim (\mathbf{1}, \mathbf{2})_{-1/2}, \quad e_R \sim (\mathbf{1}, \mathbf{1})_{-1}, \quad H \sim (\mathbf{1}, \mathbf{2})_{1/2}.$$

The core target contains no right-handed neutrino. The symbol u_R denotes the color-triplet up-type singlet and must not be replaced by ν_R . Applying $Q = T_3 + Y$ gives the point-particle charge ratios in table 4.

Table 4: Electromagnetic charge-ratio ledger in units fixed by $Q(e_R) = -1$.

Field/sector	(T_3, Y)	$Q = T_3 + Y$
$Q_L = (u_L, d_L)$	$(\pm \frac{1}{2}, \frac{1}{6})$	$(\frac{2}{3}, -\frac{1}{3})$
u_R	$(0, \frac{2}{3})$	$\frac{2}{3}$
d_R	$(0, -\frac{1}{3})$	$-\frac{1}{3}$
$L_L = (\nu_L, e_L)$	$(\pm \frac{1}{2}, -\frac{1}{2})$	$(0, -1)$
e_R	$(0, -1)$	-1
$H = (H^+, H^0)$	$(\pm \frac{1}{2}, \frac{1}{2})$	$(1, 0)$
$W^\pm$	adjoint commutator	± 1
A, Z	neutral generators	0

7.3 Gauge-boson charge

Because the Abelian generator commutes with $SU(2)$ and

$$[T_3, T^\pm] = \pm T^\pm,$$

we have

$$[Q, T^\pm] = \pm T^\pm. \quad (17)$$

Thus the charged vector directions carry charges ± 1 , while the neutral generators commute with Q . The conclusion is representation-theoretic and is independent of the neutral field basis.

Theorem 7.2 (Residual charge action and ratio rigidity; SYM8-Q05). *At D_{EW}^{1H} , the connected stabilizer line, neutral response kernel, and inherited matter representation determine one electromagnetic charge-ratio ledger up to a common nonzero unit and overall sign. Fixing $Q(e_R) = -1$ yields table 4. No weak-basis transformation, gauge choice, admissible hypercharge normalization, or neutral-field rotation changes those ratios.*

Proof. Theorem 5.1 fixes the generator line; theorem 2.4 removes Abelian normalization as a physical degree of freedom; applying the generator to each inherited irreducible representation fixes all eigenvalue ratios. Weak-basis transformations act among identical gauge representations and commute with the gauge-charge action, so they cannot alter the eigenvalues. Gauge choices change representatives, not the representation map. Neutral-field rotations transport the current pairing as in theorem 6.4. The remaining common scale and sign are fixed by the conventional electron unit and orientation. $\square$

Corollary 7.3 (Yukawa neutrality). *The inherited renormalizable Yukawa monomials are neutral under Q . This is a consistency consequence of the charge ledger and the fact that the neutral Higgs component occupies the fixation direction; it is not an independent selector of Q .*

7.4 Normalization skin and physical charge

An Abelian generator can be multiplied by λ while the coupling is divided by λ . That freedom is genuine but representational. It does not imply that charge ratios are arbitrary. The physical content lies in the pairing eQ , the ratios of eigenvalues, the action on states and local operators, and the boundary charge where admitted. A normalization convention becomes physical only after a unit is tied to an operational charge measurement; it does not generate the pattern.

Excluded alternative: u_R/v_R field substitution

The project validator treats the minimal singlet up-quark field u_R and a right-handed neutrino ν_R as different registry objects. Replacing one by the other changes color representation, hypercharge, Yukawa grammar, anomaly contributions, and the target. Such a substitution cannot pass as typography or a basis change.

Exact nonclaim boundary

The theorem fixes the charge ratios inside the inherited representation target. It does not derive the entire representation pattern from the Higgs stabilizer, determine the numerical value of e , establish electric-charge quantization for every global-form/monopole completion, or prove the exact infrared charged-state spectrum.

8 Residual electromagnetism: proper-local and boundary loci

8.1 One name, two quotient statuses

The phrase “residual $U(1)_{em}$ ” is incomplete unless the action locus is specified. A compact-support local transformation generated by a function $\alpha(x)Q$ that vanishes at the admitted boundary is a gauge operation on representatives. A transformation whose parameter approaches a nonzero value at infinity, or acts nontrivially on admitted boundary data, can carry a differentiable surface charge and act on physical charged sectors. The same Lie-algebra direction therefore supports different records at different loci without contradiction.

Let δ_α denote the infinitesimal electromagnetic gauge variation. In a canonical description its generator has the schematic form

$$G[\alpha] = \int_\Sigma \alpha \mathcal{G}_Q + Q_\partial[\alpha], \quad (18)$$

where $\mathcal{G}_Q$ is the Gauss constraint and Q_∂ is the boundary term required for differentiability. When $\alpha|_{\partial\Sigma} = 0$, the surface term vanishes and the generator is a constraint on physical states. When the boundary value is admitted and $Q_\partial \neq 0$, the transformation can act nontrivially on the physical quotient.

8.2 Proper-bulk result

Proposition 8.1 (Proper-local electromagnetic operation). *At $D_{8,\text{bulk}}$, a compact-support electromagnetic gauge transformation preserving the fixed boundary and topological data is quotient-trivial:*

$$q_{\text{proper}}(e^{i\alpha Q} \cdot x) = q_{\text{proper}}(x).$$

Its role as an operation is representative skin, while the electromagnetic connection, curvature, Gauss constraint, massless response direction, and charge pairing remain structural or physical data.

Proof. The operation is an element of $\mathcal{G}_{0,\partial}$, so [theorem 4.2](#) applies. The second sentence follows because quotient-triviality of an operation does not remove the representation, covariant derivative, response kernel, or gauge-invariant observables licensed by the gauge architecture. $\square$

8.3 Boundary and asymptotic result

Proposition 8.2 (Boundary / asymptotic electromagnetic action). *At $D_{8,\partial}$, an electromagnetic transformation receives a physical-action tag only if the target admits its boundary behavior and a quotient-level certificate shows that the differentiable charge $Q_\partial[\alpha]$ is nonzero on at least one physical sector. In that case the transformation acts on charged states or boundary observables and is not identified with the proper-local operation.*

Proof. A nonzero differentiable surface charge changes the value of a physical boundary observable or labels a charged sector. Such a transformation therefore fails the quotient-trivial criterion. Conversely, a nonvanishing parameter without a well-defined nonzero charge is insufficient; the physical-action tag requires the certificate, not the parameter name. $\square$

Theorem 8.3 (Residual electromagnetic locus split; SYM8-Q06). *The residual electromagnetic packet at D_{EW}^{1H} is the typed product*

$$\mathcal{A}_{\text{em}} = (\mathcal{A}_{\text{proper}}^{\text{trivial}}, \mathcal{A}_{\partial}^{\text{conditional physical}}, C_Q, \mathcal{F}_{\mu\nu}^{\text{em}}, Q_{\text{phys}}),$$

where proper local operations are quotient-trivial, boundary/asymptotic operations are physical exactly when certified, and the electromagnetic architecture and charge ledger remain load-bearing in both records. No single unqualified status may replace this product.

Proof. Combine [theorems 8.1](#) and [8.2](#). The two records have different target loci and quotient results, so they cannot be merged into a single bit. The common generator line and architecture explain why the same notation occurs, while the boundary certificate determines the physical action. $\square$

8.4 Charged sectors and superselection

In perturbative scattering language, Q grades matter and vector fields and constrains allowed vertices. In a more exact infrared treatment of a theory with a massless photon, charged states require attention to Gauss-law tails, soft radiation, and dressing. The present endpoint does not construct the complete infraparticle Hilbert space. Its physical charge claim is the local/operator and asymptotic charge action certified at the declared perturbative and boundary scope. The massless electromagnetic field and point-particle charge ledger are real physical structure even though the construction of exact charged asymptotic states is more subtle than a free Fock-space label.

8.5 Why this is not “leftover notation”

The following data are invariant under lawful presentations:

- the generator line that annihilates the fixation class;
- the one-dimensional kernel of the neutral response form;
- the charge ratios on the inherited representations;
- electromagnetic Ward identities and selection rules;
- the massless field-strength sector;
- the existence or absence of a certified boundary charge in a given locus.

A notation-only residue would not constrain all these objects simultaneously. The residual action theorem is therefore a positive physical theorem with a locus-qualified operation status.

Two forbidden summaries

The paper rejects both “all residual electromagnetic transformations are physical global symmetries” and “residual electromagnetism is pure gauge notation.” The first ignores the proper-local quotient; the second ignores charge, curvature, the massless response kernel, and certified asymptotic action.

Exact nonclaim boundary

The boundary theorem is schema-level until a specific asymptotic or finite-boundary phase space, falloff condition, and differentiable charge are supplied. The core endpoint asserts the locus rule and the point-particle charge action; it does not claim one universal boundary algebra for every spacetime or infrared completion.

9 Gauge-invariant physical spectrum and response-degree count

9.1 What the spectrum claim must mean

A list of gauge-fixed Lagrangian fields is not yet a physical spectrum. A reference-grade spectrum statement must identify a physical state, a gauge-invariant operator channel, an S-matrix pole, a gauge-independent real or complex pole, or a BRST cohomology class. The appropriate certificate depends on the excitation and regime. This paper therefore separates two inputs that are often conflated:

1. the *algebraic response packet*, derived locally from the Higgs representation and nonzero couplings; and
2. the *perturbative spectral certificate bundle* $\mathcal{S}_{\text{pert}}$, fixed as target data in [theorem 2.2](#).

The response calculation fixes rank, kernel, neutral directions, and charge labels. It does not by itself prove the existence of isolated physical poles. The bundle $\mathcal{S}_{\text{pert}}$ supplies the dynamical existence and gauge-transport premises needed for the physical spectrum statement.

Definition 9.1 (Certificate-conditional perturbative physical spectrum packet). Given $\mathcal{S}_{\text{pert}}$, the packet $\mathcal{P}_{\text{phys}}$ contains:

1. the admitted massless electromagnetic transverse field-strength pole sector associated with the response kernel;
2. the admitted charged and neutral massive-vector real or complex pole classes, with physical transverse and longitudinal polarizations represented in BRST cohomology and physical amplitudes;
3. one admitted neutral gauge-invariant scalar correlator/pole channel, supported by a nondegenerate radial Hessian on the selected branch, with examples such as $H^\dagger H$;
4. the inherited charged matter sectors and their Q -eigenvalues, subject to the perturbative and infrared scope stated in [section 8](#);

5. no independent physical ghost, gauge-fixing, or Goldstone-coordinate occupant.

The word “admitted” records target certification. The theorem below reconstructs the unique organization and lawful presentation orbit of these sectors; it does not derive the bundle from the response matrix.

9.2 BRST and cohomological criterion

In a perturbative gauge-fixed formulation, physical states are represented by the cohomology of the nilpotent BRST charge, schematically

$$\mathcal{H}_{\text{phys}} \simeq \ker Q_{\text{BRST}} / \text{im } Q_{\text{BRST}}.$$

Gauge, ghost, and would-be Goldstone coordinates can occur in intermediate expressions without defining additional cohomology classes. The quartet mechanism and related constructions provide the conventional basis for the cancellation of unphysical degrees of freedom [19, 20].

The cohomological statement is not used as a substitute for dynamics. It tells us how to avoid counting gauge-fixing coordinates as physical states. Pole positions, residues in physical amplitudes, and gauge-invariant correlators provide the additional dynamical certificates carried by $\mathcal{S}_{\text{pert}}$.

9.3 Gauge-parameter independence of poles

Gauge-fixed propagators can depend on ξ , and individual off-shell Green functions generally do. Nielsen identities control this dependence and imply gauge independence of physical pole positions under the assumptions of the formalism. For unstable particles the complex pole, rather than a gauge-dependent real “on-shell” parameter, is the appropriate invariant object [22, 23].

Proposition 9.2 (Gauge-choice covariance of the admitted pole packet). *Let two gauge-fixing choices lie in the same admissible perturbative chart and satisfy the BRST and Nielsen-identity hypotheses included in $\mathcal{S}_{\text{pert}}$. If their physical real or complex poles are connected by the corresponding gauge-parameter transport, then they represent one entry of $\mathcal{P}_{\text{phys}}$, even when off-shell propagators and field residues differ.*

9.4 Gauge-invariant operator channels

Useful local gauge-invariant operators include

$$O_H = H^\dagger H, \quad O_\mu = H^\dagger \overleftrightarrow{D}_\mu H,$$

and custodially organized variants. Expanding O_H around a convenient representative produces a leading term proportional to the radial coordinate, explaining why the gauge-invariant scalar channel and the familiar Higgs propagator can share a pole in the weakly coupled regime. Fröhlich–Morchio–Strocchi-type expansions sharpen this relation in suitable gauge–Higgs systems [27, 28].

The expansion is a bridge, not an identity of bare fields. The operator $H^\dagger H$ is gauge invariant; the elementary radial coordinate is gauge dependent. Their correlators can have related leading pole structure without being identical functions in all regimes. The existence of the admitted scalar channel is a clause of $\mathcal{S}_{\text{pert}}$; the expansion identifies its relation to the familiar perturbative coordinate.

9.5 Response directions and physical polarizations

The rank-three response form identifies three gauge directions that acquire nonzero quadratic response. Conditional on the vector entries of $\mathcal{S}_{\text{pert}}$, BRST and pole certificates identify those directions with three massive vector sectors, each with three physical polarizations. Conditional on the electromagnetic entry, the kernel direction is represented by one massless vector sector with two physical transverse polarizations. Conditional on the scalar entry, one neutral radial scalar channel is present. The bosonic response/polarization count is then

$$3 \times 3 + 1 \times 2 + 1 = 12,$$

which matches the coordinate reorganization described in [section 6](#). This equality is a compatibility check after the spectral premises are fixed, not an existence proof for the poles.

Theorem 9.3 (Certificate-conditional perturbative physical-spectrum reconstruction; SYM8-Q07). *Fix D_{EW}^{1H} including the spectral bundle $\mathcal{S}_{\text{pert}}$. Then the quotient-stable perturbative bosonic packet is organized as one massless electromagnetic vector sector, three massive vector sectors, and one neutral gauge-invariant scalar response/pole channel. The inherited matter sectors carry the charge ratios of [theorem 7.2](#). Gauge-fixing, ghost, and would-be Goldstone coordinates do not add independent physical occupants. The packet is invariant under admitted gauge-fixing and field-basis changes when expressed through BRST cohomology, physical amplitudes, gauge-invariant correlators, and the real or complex pole classes certified by $\mathcal{S}_{\text{pert}}$.*

Proof. The response theorems SYM8-Q03 and SYM8-Q04 supply one null and three nonnull gauge directions and fix their electromagnetic, charged, and neutral labels. They do not prove pole existence. The bundle $\mathcal{S}_{\text{pert}}$ independently supplies a nilpotent BRST complex, the admitted photon field-strength pole, admitted $W^\pm$ and Z real or complex pole classes, and an admitted neutral scalar invariant-correlator/pole channel with nondegenerate radial Hessian. In the resulting perturbative BRST description, unphysical gauge, ghost, and angular coordinates are absent from cohomology or cancel in physical amplitudes; the nonzero response directions organize the three admitted massive-vector classes, while the kernel organizes the admitted electromagnetic class. The scalar certificate identifies the neutral scalar channel, and [theorem 7.2](#) fixes the matter and vector charge labels. Gauge-parameter and field-basis transports preserve the physical cohomology, amplitudes, invariant correlators, and admitted pole classes by the certificates in $\mathcal{S}_{\text{pert}}$. Hence every admitted presentation reconstructs the same physical packet. $\square$

9.6 Unstable particles and exact language

The W , Z , and Higgs are unstable in the observed theory. The manuscript therefore uses “sector,” “response direction,” and “pole class” where an exact stable one-particle state would be misleading. A complex pole and its physical residue structure can still organize scattering and resonance phenomena. This vocabulary strengthens, rather than weakens, the gauge-invariant conclusion.

9.7 Sector-by-sector certificate map

The normal form does not use one universal notion of “particle.” Each sector carries the strongest certificate appropriate to its physics.

9.7.1 Electromagnetic sector

The algebraic certificate is the unique null direction of the response form and the universal coupling eQ . The perturbative dynamical certificate, explicitly included in $\mathcal{S}_{\text{pert}}$, is a massless transverse pole in gauge-invariant electromagnetic field-strength correlations and physical amplitudes. The asymptotic certificate is the action of the electric charge where the boundary phase space admits it. A complete infrared charged-state construction is not required to identify the massless field-strength sector.

9.7.2 Charged massive-vector sectors

The charged response directions are fixed by $T^\pm$ and carry $Q = \pm 1$. In a renormalizable gauge their propagators, Goldstone mixing, and ghosts are gauge dependent off shell, but physical amplitudes and the admitted complex pole classes are gauge independent under the standard identities. The longitudinal mode is not a fourth polarization; it is one of the three physical polarizations of a massive spin-one response sector. The Goldstone representative used at high energy is linked by [theorem 10.2](#).

9.7.3 Neutral massive-vector sector

The Z direction is the unique neutral image direction orthogonal to the photon with respect to the canonical kinetic metric. Its coupling is the transported current $g_2(T_3 - s_W^2 Q)/c_W$. This pairing, not the letter Z , identifies the sector. A neutral basis that swaps labels or changes signs is lawful only if the response, current, and pole maps are transported together.

9.7.4 Scalar sector

The radial coordinate h is a useful perturbative field, while $H^\dagger H$ is a local gauge-invariant scalar operator. Expanding in a convenient chart gives

$$H^\dagger H = \frac{1}{2}(v + h)^2 + \text{angular-chart terms arranged gauge invariantly},$$

so the connected correlator contains a leading contribution proportional to the radial propagator. The physical claim is the target-certified existence of a neutral scalar response/pole channel in the selected regime, not literal identity of the elementary and composite fields.

9.7.5 Fermionic sectors

Chiral matter fields are gauge covariant and carry the charge ratios fixed in [table 4](#). In perturbative scattering they label charged external sectors after the usual dressing and infrared qualifications. Mass eigenvalues and flavor mixing are downstream of Yukawa tensors and weak-basis quotienting; their numerical values are not determined here. Electric charge commutes with lawful flavor-basis transformations, so the charge ledger is generation independent even when the mass basis changes.

9.8 Pole, residue, and state distinctions

A pole position, residue, and asymptotic state are related but not identical concepts. Stable particles can correspond to isolated real poles and one-particle states. Unstable electroweak resonances are more properly described by complex poles and physical scattering residues. Gauge-fixed propagator residues can be gauge dependent even when a complex pole is not. The spectrum packet therefore stores:

1. the channel or representation label;
2. the physical pole class, real or complex as appropriate;
3. the amplitude or gauge-invariant correlator in which it is detected;
4. the gauge-parameter transport certificate;
5. the stability and infrared qualification.

A submission that supplies only a field name or a gauge-dependent propagator parameter is certificate-incomplete. This discipline is essential to the normal-form theorem: apparent extra “particles” generated by gauge choices disappear when the physical certificate is demanded, while a genuinely new certified pole or state would constitute new same-target physical content.

9.9 Why the response count and spectrum packet are both needed

The response rank is an algebraic fact and is insensitive to many dynamical details. It ensures that three gauge directions receive nonzero quadratic response and one does not. It does not by itself prove that every response direction produces an isolated physical pole in every regime. Conversely, a numerical spectrum list without the response map would not explain why the massless and massive directions are linked to the stabilizer and charge generator. The endpoint therefore contains both objects and a compatibility condition between them.

At the selected target, $\mathcal{S}_{\text{pert}}$ supplies the admitted perturbative sectors, and the local response theorem proves their unique organization: the three nonzero response directions correspond to the $W^\pm$ and Z pole classes, while the kernel corresponds to the massless photon sector. A more strongly coupled

target might require different gauge-invariant composite labels and could alter spectral details without changing the local response algebra. Such a target would need a new spectral bundle and cannot be silently substituted for $\mathcal{R}_{\text{pert}}$.

9.10 Nonperturbative boundary

A fully nonperturbative theorem would have to specify the operator algebra, state construction, phase, lattice or continuum definition, faithful global form where needed, and infrared charged sectors. It might use gauge-invariant composite operators rather than elementary fields as its primary spectrum labels. The local normal-form theorem does not require that stronger result. It requires that every admitted perturbative presentation preserve the same target-certified physical packet under the declared QFT hypotheses.

Exact nonclaim boundary

No universal theorem is asserted that every elementary electroweak field has an exactly identical local gauge-invariant composite counterpart with the same full correlator. The claim is limited to the gauge-invariant or gauge-independent channel and pole reconstruction licensed by $\mathcal{S}_{\text{pert}}$ in the selected weakly coupled phase.

10 Goldstone-language translation and longitudinal vector content

10.1 Angular coordinates in a gauge-fixed chart

Near a nonzero representative one may write, schematically,

$$H(x) = \frac{1}{\sqrt{2}} \exp\left(\frac{i\pi^a(x)\sigma^a}{v}\right) \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}, \quad (19)$$

within a local chart. The fields π^a are angular coordinates along gauge directions. Different gauges distribute them differently among scalar, gauge, ghost, and mixing terms. In unitary gauge they are absent from the displayed scalar multiplet; in R_ξ gauges they appear with gauge-dependent masses and propagate in intermediate diagrams.

The coordinate role is real and useful. It is not an independent physical particle count at the core gauge locus. The physical longitudinal polarization of a massive vector is represented by a cohomology class and scattering amplitudes; the π^a fields are calculational representatives connected to that content.

10.2 BRST status and double-counting prohibition

In a renormalizable gauge, the gauge-fixing function typically couples $\partial \cdot V^a$ to π^a . The unphysical scalar, longitudinal gauge, and ghost coordinates belong to the BRST complex. Counting the three π^a fields in addition to the three longitudinal vector polarizations would count one construction twice.

Proposition 10.1 (No independent Goldstone occupant at the gauge locus). *At D_{EW}^{1H} , the gauge-fixed would-be Goldstone coordinates π^a do not define three additional physical particle sectors beyond the longitudinal polarizations of the massive electroweak vectors. A candidate that counts both fails the physical cohomology and degree-count certificates.*

10.3 The high-energy amplitude theorem

For external longitudinal vector bosons with characteristic energy $E \gg m_V$, the equivalence theorem has the schematic renormalized form

$$\mathcal{A}(V_L^{a_1}, \dots, V_L^{a_n}; X) = \left(\prod_{k=1}^n C_{a_k} \right) \mathcal{A}(\phi^{a_1}, \dots, \phi^{a_n}; X) + \mathcal{O}(m_V/E), \quad (20)$$

where ϕ^a is the corresponding gauge-fixed Goldstone field and C_a is a finite normalization factor determined by the renormalization convention and mixing structure [24, 25, 26]. The equality concerns amplitudes in a stated regime, not equality of local fields or exact state identities.

Theorem 10.2 (Goldstone/longitudinal translation; SYM8-Q08). *The SYM-8 Goldstone packet is a typed translation*

$$\mathcal{T}_{G/L} = (\mathcal{R}_{\text{ang}}, C_{\text{BRST}}, \mathcal{P}_L; C_a, E/m_V, \mathcal{R}_{m/E}),$$

where $\mathcal{R}_{\text{ang}}$ is the gauge-fixed angular chart, C_{BRST} its cohomological status, $\mathcal{P}_L$ the longitudinal physical sector, and $\mathcal{R}_{m/E}$ the correction certificate. At the gauge locus, ϕ^a is a representative used to compute longitudinal amplitudes and does not add an independent physical occupant. In the high-energy regime the amplitudes satisfy (20). Outside that regime, the correction terms must be retained and no exact replacement is inferred.

Proof. The nonoccupancy clause follows from theorem 10.1 and the BRST physical-state criterion. The amplitude clause is the imported equivalence theorem under its stated kinematic and renormalization hypotheses. The packet records the finite factor and remainder, so the relation is transported without promoting it to a field identity or all-energy theorem. $\square$

10.4 Gauge and global target split

The Goldstone theorem for a spontaneously broken global continuous symmetry concerns a physical current, a vacuum not invariant under a physical action, and massless excitations under its assumptions. The electroweak would-be Goldstone coordinates at the proper local gauge locus do not satisfy the same quotient structure. In a controlled global custodial or chiral target, physical Goldstone modes can occur. The word “Goldstone” therefore requires a target and action-locus tag.

Table 5: Gauge-locus and global-locus Goldstone statements.

Field	Gauge-locus statement	Global-locus statement
Action	Proper local redundancy, quotient-trivial	Physical global action on states/operators
Vacuum	Gauge-related orientations one quotient class	Vacuum may fail to be invariant physically
Mode	Angular gauge coordinate; longitudinal representative	Physical massless Goldstone mode when assumptions hold
Mass	Gauge-dependent coordinate mass in R_ξ gauge	Protected/controlled by global symmetry realization
Translation	High-energy amplitude relation with corrections	Nonlinear realization and current algebra

10.5 What “eaten” preserves

The shorthand “the vector eats the Goldstone” preserves a degree-of-freedom reorganization: a massless vector with two polarizations and an angular scalar coordinate are represented after fixation by a massive

vector with three physical polarizations. It does not mean that an independently standing physical particle is consumed by another particle. The phrase is admissible when accompanied by this translation.

10.6 Failure modes

The following mutations are explicitly rejected:

1. replacing $O(m_V/E)$ by zero at arbitrary energy;
2. dropping the finite factors C_a beyond a convention where they are unity;
3. using an off-shell Green-function relation as an on-shell S-matrix theorem;
4. counting ϕ^a and V_L^a as independent physical sectors;
5. transferring the gauge-locus conclusion to a global-symmetry target.

Exact nonclaim boundary

The theorem does not deny the calculational importance of Goldstone fields, ghost fields, or R_ξ gauges. It denies only their promotion into extra physical occupants or an exact all-energy field identity at the core target.

III

Structural integration

11 Higgs bridge/fixation compatibility and the no-promotion theorem

11.1 Imported Higgs endpoint

The Higgs Structural Endpoint Arc fixes a standing-bearing carrier

$$C_H = H_{\text{br/fix}} \quad (21)$$

for the same-scope order-to-deformation and mass-response role. It further fixes Higgs structural identity as admissible occupation of that carrier and proves that field, vev, potential minimum, pole mass, running coordinate, scale, vacuum branch, and extension descriptions do not supply independent primitive identity grounds [7, 8, 9, 10, 11, 12]. In compact form,

$$\text{Id}_H = \text{Occ}_{D_H}(C_H), \quad \text{Surv}_{D_H}^{\text{prim}}(\text{Attr}_H \setminus C_H) = 0. \quad (22)$$

These are imported endpoint statements. SYM-8 uses them to type the electroweak fixation and response construction; it does not use them to prove the QFT stabilizer or response matrix.

11.2 The local bridge map

The one-doublet electroweak construction defines a local map

$$\mathfrak{B}_8 : ([H_0]_{\mathcal{G}_{0,\partial}}, \rho_H, D_\mu) \mapsto (\text{Ann}(H_0), \mu_{H_0}, \mathcal{P}_{\text{phys}}), \quad (23)$$

followed by the charge and locus maps

$$(\text{Ann}(H_0), \mu_{H_0}) \mapsto (Q, Q_{\text{phys}}, \mathcal{A}_{\text{em}}).$$

The domain includes the fixation class and inherited gauge representation. The codomain contains the stabilizer, response, spectrum, charge, and action packets. Under a lawful presentation arrow, the domain and codomain are transported together. The map is therefore a factorization through the Higgs carrier role, not a declaration that the literal coordinate H_0 is primitive.

11.3 Representative and response catalogue

The following entries remain indispensable to calculations but have different structural statuses.

Table 6: Higgs-associated entries in the SYM-8 construction.

Entry	Raw physical role	SYM-8 standing status
Scalar doublet H	Local field representative and carrier realization	Licensed representative of C_H , not a second identity ground
Gauge-fixed H_0	Convenient fixation coordinate	Representative of $[H_0]_{\mathcal{G}_{0,\partial}}$
v	Response coordinate/-electroweak input	Carrier-indexed scale representative; not primitive scale
Potential chart $V(H)$	Encodes local interactions and branch structure	Downstream description/route; not residual-selector authority
Radial field h	Perturbative coordinate for scalar channel	Representative connected to a gauge-invariant/pole channel
Pole mass/width	Physical response measurement	Downstream spectrum witness; not Higgs identity
Goldstone coordinates ϕ^a	Longitudinal calculation representatives	Translation coordinates; no extra occupant
Custodial bidoublet Φ	Invertible repackaging of one doublet	Same carrier, enlarged display for target-limit symmetry

Theorem 11.1 (Higgs bridge compatibility and no promotion; SYM8-A01). *Every same-target Higgs-associated entry used in the SYM-8 normal form factors through the inherited bridge/fixation carrier C_H and the local map (23). No field coordinate, vev, potential chart, radial coordinate, Goldstone chart, pole datum, gauge choice, or custodial repackaging acquires independent selector authority for the vacuum, stabilizer, charge ledger, or spectrum. A submitted Higgs object that does not preserve C_H , the one-doublet target, and the quotient certificates is either representative-only, role-incomplete, or a target update.*

Proof. The local QFT construction uses the Higgs representation and fixation class to produce response and residual data. By the imported Higgs endpoint (22), any same-scope identity-bearing use must occupy C_H , while the listed coordinates and measurements are representatives or downstream routes. The normal-form compatibility predicate also requires preservation of the fixation orbit, response rank/kernel, charge pairing, physical spectrum, and target. Therefore a coordinate can be changed lawfully only when those data are transported. If it is claimed to select them independently, the claim reverses the established dependency and fails the carrier or quotient certificate. If the object adds new scalar content or changes the carrier role, it leaves D_{EW}^{1H} . $\square$

Corollary 11.2 (No vev selector). *The value and orientation of a gauge-fixed vev cannot, by themselves, determine Higgs identity or physical electromagnetism. The value enters the response scale, while the representation and fixation class determine the annihilator and rank structure.*

Corollary 11.3 (No pole selector). *A measured scalar pole is a physical response witness in the admitted channel, but it does not ground the Higgs carrier or the electroweak residual generator. The carrier and representation are logically upstream of the pole assignment.*

11.4 Vacuum route status

The Higgs vacuum-route manuscript distinguishes route availability from standing closure:

$$\text{RouteAvail}_{D_H}(v) \not\Rightarrow \text{StandingClosed}_{D_H}(v).$$

SYM-8 uses only the selected local nonzero fixation branch. It does not infer global vacuum stability, metastability, tunneling lifetime, thermal history, or baryogenesis. A lower branch or alternative thermal phase may be physically meaningful in a richer target while remaining downstream of the same carrier or defining a target update.

11.5 No dependency inversion

The Higgs endpoint does not prove the electroweak response rank by structural fiat. The rank is locally computed from the representation and kinetic term in [section 6](#). Conversely, the successful computation does not replace the Higgs identity theorem with the statement “whatever gives this mass matrix is the Higgs.” The two results have different burdens and compose in a fixed order:

Higgs carrier and identity $\longrightarrow$ one-doublet fixation representation
 $\longrightarrow$ response and residual packet $\longrightarrow$ downstream measurements.

Same response does not automatically mean same carrier

A BSM scalar sector may reproduce some tree-level masses while changing field content, custodial structure, couplings, higher operators, global form, or spectrum. Such a construction is not a same-target rival merely because one matrix agrees. It must either exhibit a carrier-preserving equivalence of the full target or be registered as a target update.

Exact nonclaim boundary

The theorem does not deny scalar-field ontology, experimental Higgs identification, or the utility of the potential and vev. It denies only their promotion to an additional primitive identity or residual-sector selector independent of the bridge/fixation carrier and physical certificates.

12 Custodial interface and electromagnetic noncollapse

12.1 One doublet, two useful presentations

Define

$$\tilde{H} = i\sigma_2 H^*, \quad \Phi = (\tilde{H}, H).$$

For one Higgs doublet, Φ is an invertible repackaging of the same four real coordinates. It does not add a second scalar carrier. In the appropriate scalar-limit presentation, the potential can be organized by a left–right action

$$\Phi \mapsto U_L \Phi U_R^\dagger, \quad (U_L, U_R) \in SU(2)_L \times SU(2)_R,$$

with the common central $\mathbb{Z}_2$ acting trivially, so the faithful scalar action is

$$[SU(2)_L \times SU(2)_R]/\mathbb{Z}_2 \simeq SO(4).$$

This is an imported SYM-6 interface [\[5\]](#).

Table 7: Custodial target packet imported from SYM-6.

Target	Exact/defining conditions	Retained qualification
Scalar target	One-doublet scalar kinetic/potential structure in the custodial form	No claim about full gauge and matter sectors
Gauge–Higgs target	$g_1 = 0$ with the one-doublet scalar sector	Hypercharge gauging is a distinct deformation
Minimal-matter target	$g_1 = 0$, aligned quark Yukawa tensors, lepton-sector qualification	Missing ν_R obstructs a naive full right doublet
Dirac-neutrino target	Add ν_R , set $g_1 = 0$, impose quark and lepton Yukawa alignments	Explicit target update with its own flavor packet

12.2 Target-indexed custodial packet

Custodial symmetry is not one unqualified property of the observed Standard Model. The upstream packet distinguishes at least four targets.

For a representative $\langle \Phi \rangle = vI/\sqrt{2}$, the stabilizer in the covering group is the diagonal subgroup

$$SU(2)_V = \{(U, U)\}.$$

Because the common center has already been divided out in the faithful scalar action, the faithful stabilizer image is

$$SU(2)_V/\{\pm I\} \simeq SO(3)_V.$$

The conventional label “custodial $SU(2)_V$ ” refers to this diagonal cover unless the faithful quotient is being discussed explicitly. The roles must remain separated: the left factor overlaps with gauged electroweak architecture; the right factor is global in the target; and the diagonal image organizes physical responses only in the appropriate exact or approximate limit.

12.3 Deformation ancestry

The custodial packet records distinct sources of departure rather than one generic “breaking.” Relevant coordinates include:

- hypercharge gauging $g_1 \neq 0$;
- up/down Yukawa splittings, including the large top/bottom difference;
- lepton and neutrino field content and Yukawa structure;
- the absence of a right-handed neutrino in the minimal target;
- higher-dimensional operators;
- renormalization-group transport and loop effects.

Each source has a different target and tensor ancestry. A diagnostic such as $\rho = 1$ or $T = 0$ may follow in a limit but is not equivalent to the full custodial symmetry statement.

12.4 Why custodial and electromagnetic stabilizers differ

The connected electromagnetic stabilizer algebra $\mathfrak{u}(1)_Q$ is the annihilator inside the gauged electroweak algebra at the physical nonzero branch. Its proper-local operations remain gauge redundancy, while its global/asymptotic charge action grades physical sectors. The custodial packet instead comes from the faithful global left–right action $SO(4)$, with diagonal faithful stabilizer image $SO(3)_V$, and is available

only in controlled scalar/gauge/Yukawa limits. Their dimensions, actions, exactness conditions, deformation ancestry, and quotient loci differ.

Theorem 12.1 (Electromagnetic/custodial noncollapse; SYM8-A02). *At D_{EW}^{1H} , the electromagnetic residual packet $\mathcal{A}_{em}$ and the custodial packet $\mathcal{C}_{cust}$ are not isomorphic symmetry records:*

$$\mathcal{A}_{em} \neq \mathcal{C}_{cust}, \quad \dim \mathfrak{u}(1)_Q = 1, \quad \dim \mathfrak{so}(3)_V = 3.$$

The first is fixed by the one-dimensional annihilator and charge action of the electroweak gauge representation. The second is an exact or approximate global/limit symmetry whose diagonal cover is conventionally denoted $SU(2)_V$ and whose faithful stabilizer image is $SO(3)_V$ under the conditions of table 7. An equality of selected mass relations or a common use of the word “stabilizer” does not identify them.

Proof. The electromagnetic record has generator algebra $\mathfrak{u}(1)_Q$, acts with the charge ratios of table 4, and is exact as a gauge architecture at the core target with locus-split operation status. The custodial record has faithful diagonal stabilizer $SO(3)_V$ (or diagonal cover $SU(2)_V$), requires the target restrictions in table 7, and is deformed by nonzero hypercharge and generic Yukawa data. The records have different dimensions and different actions on the fields. Their status vectors therefore differ in acting object, target, quotient locus, exactness, and deformation fields. They cannot be one record under the SYM-0 calculus. $\square$

Corollary 12.2 (Diagnostic firewall). *The tree-level relation $\rho = 1$, a small oblique T parameter, or an approximate vector-mass pattern can diagnose custodial protection in a declared regime but cannot define custodial symmetry or prove the full custodial target conditions.*

12.5 The physical core and the custodial limit

The observed one-doublet electroweak theory can be close to custodial in selected observables while not possessing exact full custodial symmetry. The endpoint therefore retains the packet as an interface rather than deleting it. This illustrates a central arc principle: an approximate or limit symmetry may be physically powerful without being primitive or exact in the full target.

Excluded alternative: diagonal subgroup as residual gauge group

Replacing the electromagnetic generator by a generator of the custodial diagonal cover (or its faithful $SO(3)_V$ image) fails immediately: the custodial action is not an alternative one-dimensional annihilator inside the physical gauged algebra, does not reproduce the electric-charge ledger, and requires a different target and acting object. It is not a second residual electromagnetic solution.

Exact nonclaim boundary

No numerical electroweak precision fit is performed, and no claim is made that custodial symmetry is exact in the observed Standard Model. The paper imports the exact target packet and its controlled deformation map only.

13 Phase versus symmetry: invariant signatures and gauge-fixed charts

13.1 Why a gauge chart is not a phase

Two gauge choices can make the same physical electroweak regime look different. In one chart the scalar expectation value is explicit; in another a local order parameter may vanish; in unitary gauge angular coordinates disappear; in an R_ξ gauge they propagate. A phase statement that changes with these choices is not yet physical. Conversely, the absence of a local gauge-variant order parameter does not imply that no physical response or spectrum distinction exists.

13.2 Invariant phase signature

For the limited purpose of SYM-8, define the signature

$$\Sigma_{\text{phase}} = (\dim \ker \mu, \text{rank } \mu, Q_{\text{phys}}, P_{\text{phys}}, \mathcal{A}_{\text{em}}, O_{\text{inv}}, \mathcal{B}, \mathcal{R}), \quad (24)$$

where O_{inv} denotes the admitted gauge-invariant correlation/response data, $\mathcal{B}$ the boundary/global locus, and $\mathcal{R}$ the quantum/energy regime. The signature is not claimed to classify every nonperturbative gauge–Higgs phase. It is the invariant packet required to decide whether two presentations in this manuscript describe the same selected electroweak phase.

Definition 13.1 (Same SYM-8 phase). Two presentations are in the same SYM-8 phase class when they have the same target and their signatures Σ_{phase} are connected by lawful presentation arrows preserving the listed certificates.

13.3 Gauge-fixed chart rule

Theorem 13.2 (Phase-versus-presentation rule; SYM8-Q09). *Let P and P' be two gauge-fixed or field-coordinate presentations of D_{EW}^{1H} . If a lawful arrow transports their physical quotient, response rank/kernel, charge ledger, boundary locus, physical spectrum, and invariant correlation data, then they represent the same SYM-8 phase and their differences are presentation skin. If one of those invariant fields changes, the difference is physical or a target update and cannot be dismissed as gauge choice.*

Proof. The forward direction is the definition of the lawful presentation groupoid and phase signature. For the converse, a change in any listed field changes an observable, a physical-action locus, the spectrum packet, or the target. Such a change cannot be generated by an arrow defined to preserve those data. Therefore it is not presentation-only. $\square$

13.4 Worked comparisons

13.4.1 Unitary gauge and R_ξ gauge

The displayed scalar and propagator content differs, but the response rank/kernel, charge ledger, physical pole classes, and scattering amplitudes agree under the BRST and Nielsen certificates. The two charts represent the same phase.

13.4.2 Two Higgs orientations

If $H'_0 = gH_0$ with $g \in \mathcal{G}_{0,\partial}$, then the annihilator is conjugated and the response form transported. The fixation class and all physical signature fields agree. The orientations are one phase and one physical vacuum class.

13.4.3 The $v = 0$ comparison target

At $v = 0$, the response form vanishes at tree level and the rank/kernel signature changes. This is not a gauge copy of the nonzero branch. Whether the two regimes are separated by a genuine thermodynamic or nonanalytic phase boundary in a specified nonperturbative model is a further question; the target has already changed at the fixation coordinate and response signature.

13.4.4 Added scalar multiplet

An additional scalar can alter the stabilizer, response rank, custodial structure, and physical spectrum. It is a target update even if a gauge choice makes one component resemble the one-doublet representative.

13.5 Fradkin–Shenker boundary

Gauge–Higgs lattice systems can exhibit analytic continuity between regions that naive gauge-fixed language might call “Higgs” and “confinement” [29]. This result reinforces the need for invariant

diagnostics and blocks a universal phase theorem based only on a gauge-variant vev. SYM-8 does not attempt to classify the complete phase diagram. It fixes one selected weakly coupled phase and proves presentation invariance inside it.

13.6 Vacuum degeneracy and global symmetries

A genuine global symmetry can produce physically distinct vacua when the action is nontrivial on the physical quotient and the state is not invariant. The gauge-fixed electroweak orbit does not meet that criterion at the proper local locus. A custodial or other global target must be evaluated separately with its own order parameter, current, anomaly, and vacuum certificates.

Phase certificate

Every claim that two electroweak descriptions are the same phase must provide: target identity; a forward and inverse presentation map; response-rank and kernel transport; charge-pairing preservation; pole/cohomology correspondence; boundary-locus identity; and invariant-correlation compatibility. Visual resemblance of potentials or fields is insufficient.

Exact nonclaim boundary

The theorem is not a universal classification of electroweak finite-temperature phases, lattice phases, crossover behavior, or nonperturbative phase boundaries. It is a same-target presentation theorem for the selected one-doublet perturbative phase.

14 AASC role map and the SYM-8 atlas

14.1 Role assignment after physical reconstruction

With the framework-neutral objects fixed, the SYM-0 role vocabulary can be applied without circularity. A single familiar word may generate several rows because roles are locus- and level-dependent. The atlas records the physical status first and the structural role second.

Table 8: Readable projection of the sixteen-row SYM-8 atlas.

Record	Object	Quotient action	Structural role	Evidence
SYM8-TARGET-1H-CORE-v1	one-doublet electroweak core target	not applicable	anchor/target	local declaration
SYM8-VAC-FIXATION-ORBIT-v1	nonzero fixation orbit	trivial on proper quotient	bridge/fixation witness	SYM-1 plus local proof
SYM8-GAUGE-LOCAL-NONORDER-v1	local gauge-variant vev is not a physical order parameter	trivial	skin operation; derived theorem	SYM-1, Elitzur-type source, local proof
SYM8-STAB-EM-CONNECTED-v1	connected electromagnetic stabilizer	structural; conjugacy class invariant	structural residual architecture	local proof
SYM8-RESP-RANK3-KERNEL1-v1	gauge-response rank three, kernel one	nontrivial physical response	derived witness/tensor	local proof
SYM8-NEUTRAL-AZ-NORMALFORM-v1	neutral A/Z normal form	representative basis; invariant kernel/image	skin/transport chart plus physical witness	local proof

Record	Object	Quotient action	Structural role	Evidence
SYM8-CHARGE-Q-GENERATOR-v1	electromagnetic generator line	nontrivial on charged sectors	physical action architecture; derived witness	local proof plus charge-ratio import
SYM8-CHARGE-POINT-SECTOR-v1	point-sector charge ratios	nontrivial	physical tensor/ action	generated from field registry
SYM8-EM-PROPER-LOCAL-v1	proper local electromagnetic operation	trivial	skin	SYM-1 import and specialization
SYM8-EM-BOUNDARY-ACTION-v1	boundary/asymptotic electromagnetic action	nontrivial when charge certificate exists	physical action; surface-charge witness	SYM-1 boundary theorem and specialization
SYM8-SPECTRUM-PERTURBATIVE-v1	perturbative physical response spectrum	nontrivial	physical states/ witnesses	local proof plus QFT source dossiers
SYM8-GOLDSTONE-LONGITUDINAL-v1	Goldstone-longitudinal translation	Goldstone coordinate trivial as independent cohomology; amplitude map physical	computational route plus physical response	equivalence-theorem source and local integration
SYM8-HIGGS-BRIDGE-COMPAT-v1	Higgs bridge compatibility and no promotion	representatives descend to one carrier-compatible package	bridge/fixation carrier; downstream witnesses	Higgs arc import plus local factorization
SYM8-CUSTODIAL-NONCOLLAPSE-v1	electromagnetic/custodial non-collapse	different target/action loci	physical action versus effective/limit route	SYM-6 C01-C06 and local comparison
SYM8-PHASE-REPRESENTATIVE-RULE-v1	phase versus presentation rule	trivial if signature preserved	skin classification or target update	local theorem
SYM8-NORMALFORM-RIGIDITY-v1	one-doublet electroweak normal-form rigidity	normal-form invariants unchanged under admitted presentation changes	integrated normal-form theorem	local and imported theorem composition

14.2 Multi-role compatibility

Several rows legitimately carry more than one role, but only across distinct levels.

- Electromagnetic structure is a residual gauge architecture, a physical charge action on sectors, and a set of derived Ward identities. A proper local operation remains skin.
- The response form is a physical tensor on the gauge-direction space and a witness of the fixation structure; it is not a new carrier independent of C_H .
- Custodial symmetry is a physical/effective global action in its target and a diagnostic organizer in the physical theory; it is not the electromagnetic gauge stabilizer.
- The Higgs fixation class is a bridge/fixation state, while its displayed vev is a representative coordinate.

No row is both skin and an unqualified physical action at the same quotient locus.

14.3 Stable identifiers and evidence packets

The sixteen canonical rows each have a stable identifier, target vector, acting object, acted-upon object, quotient status, anomaly status, vacuum status, boundary locus, scale/energy scope, role, proof source,

rival disposition, and nonclaim. Finer-grained objects such as the BRST complex, the gauge-fixing parameter, the individual vector directions, and the custodial diagnostics are manuscript projections of those integrated rows. They are not silently counted as additional canonical atlas records. The machine-readable release in `data/atlas.csv` and `data/atlas.json` is authoritative for the project build; the validator checks required fields, identifiers, and target joins.

14.4 Role-map covariance

Theorem 14.1 (SYM-8 role-map covariance). *If two complete SYM-8 records are related by a lawful presentation arrow, their quotient status, physical invariant content, and AASC role set agree after transport. A change in literal formula, basis, gauge, normalization, or repackaging does not create a new role.*

Proof. Lawful arrows preserve the target, quotient, anomaly, boundary, response, charge, spectrum, carrier, and phase certificates by [theorems 3.1](#) and [3.3](#). The role assignment rules depend on those certificates: removal, quotient action, physical action, dependency, and target change. Therefore the role outcome is invariant. If a role changes, at least one certificate has changed, so the arrow was not lawful. $\square$

14.5 Representative/invariant firewall

Table 9: Representative coordinates and invariant content.

Representative/route	Invariant or quotient-stable content
Higgs orientation H_0	Proper-gauge fixation class and stabilizer conjugacy class
Vev coordinate v	Response scale relative to fixed renormalized inputs
Neutral fields (W^3, B) or (A, Z)	One-dimensional kernel, one-dimensional neutral image, transported current pairing
Hypercharge normalization	Pairing $g_1 Y$, generator line, charge ratios
Weak flavor basis	Mass spectra, mixing invariants, charge action
Gauge parameter ξ	Physical pole/cohomology/S-matrix packet
Goldstone field ϕ^a	Longitudinal amplitude relation and physical vector sector
Custodial bidoublet Φ	One Higgs carrier and target-indexed global symmetry packet
Potential chart	Selected branch and invariant response/correlation data

Exact nonclaim boundary

The role map does not replace the QFT derivations in preceding chapters. A reader who rejects AASC can still evaluate the stabilizer, response, charge, spectrum, and amplitude results. The structural layer classifies their dependency and standing; it does not manufacture them.

IV

Normal-form synthesis, rival audit, and theorem boundary

15 Scope-preserving rival analysis and declared finite audit

15.1 Why a rival ledger is still required

The local QFT results establish the electroweak normal form, but a publication-grade argument should also make its likely failure modes visible. A rival ledger serves three purposes. It distinguishes a changed target from a same-target objection; it prevents a presentation difference from being mistaken for new physics; and it records which theorem or certificate bears the weight of each disposition. The ledger is therefore an audit instrument attached to the scientific proof, not an independent source of scientific truth.

The decisive question for every proposed rival is not whether it resembles the canonical display. It is whether it preserves the target and all quotient-level interfaces while changing a physically invariant entry of $\mathfrak{C}_{EW}$. A gauge-rotated Higgs orientation, rescaled generator, weak-basis change, or neutral-field coordinate change may be a lawful redescription. A second null direction, altered charge ratio, additional physical Goldstone, or custodial/electromagnetic identification conflicts with a specific theorem. An extra scalar, different neutrino content, nontrivial global-form completion, or nonperturbative charged-state construction changes the target.

15.2 Disposition classes

The rival protocol of [theorem 3.4](#) uses the following classes.

Normal-form representative

A convenient canonical display of one entry of $\mathfrak{C}_{EW}$.

Lawful redescription

A different display connected by an explicitly stated transformation that transports all dependent data.

Physically incompatible

A same-target claim that violates a proved response, charge, quotient, spectrum, carrier, custodial, or phase interface.

Certificate-incomplete

A claim for which the required physical or source certificate has not been supplied.

Controlled target fiber

A nearby parameter limit or branch tracked explicitly but not identified with the core target.

Target update

A change of fields, operators, boundary data, global form, or quantum regime.

Target mismatch

A claim that silently imports data excluded by the frozen target.

Open same-target burden

A fully specified same-target claim not resolved by the existing theorem chain.

The final class remains available even when the published audit contains no such row.

15.3 Declared registry

The published rival audit contains fifty concrete schemas distributed across the ten loci of [table 3](#). The rows include canonical representatives, lawful gauge and basis presentations, malformed transports, altered rank and charge claims, spectrum and Goldstone overcounts, Higgs-identity promotions, custodial collapses, target fibers, and explicit target updates. The readable summary is generated from `data/candidate_audit.csv`; the complete row-level file remains the authority.

Table 10: Disposition counts in the declared fifty-row rival audit.

Audit disposition	Rows
normal-form representative	10
lawful redescription	11
physically incompatible	16
certificate-incomplete	2
controlled target fiber	4
target update	6
target mismatch	1
Total declared audit rows	50

Theorem 15.1 (Declared finite rival audit; SYM8-A03). *Every row in the published fifty-schema SYM-8 rival registry has:*

1. *an explicit target relation;*
2. *a named physical locus;*
3. *a decisive theorem, imported interface, or certificate requirement;*
4. *one audit disposition from the classes above; and*
5. *an exact nonclaim preventing that row from being promoted into a universal completeness statement.*

The registry contains no unresolved internal contradiction among its declared rows.

Proof. The target relation and physical locus are recorded in the rival and candidate-audit files. Canonical representatives and lawful redescrptions are supported by the presentation-covariance lemma and the relevant component theorem. Physically incompatible rows are tied to the exact failed interface: quotient, annihilator, response rank or kernel, charge ratio, boundary action, spectrum, Goldstone regime, Higgs-carrier provenance, custodial target, or phase signature. Incomplete rows name the missing certificate. Fibers, updates, and mismatches identify the changed target field. The source and theorem ledgers provide the cited authority, and the generated table is checked against the machine-readable rows. This proves completeness of the published row audit. It does not prove that the registry contains every possible same-target proposal. $\square$

15.4 Representative examples

Table 11: Representative rows from the declared rival audit; the full fifty-row CSV is authoritative.

Audit ID	Challenge	Disposition	Decisive result	Audit basis
VAC-01	canonical nonzero fixation representative	normal-form representative	SYM8-Q01	complete canonical invariant with certified map family
VAC-02	proper-gauge rotated nonzero representative	lawful redescription	SYM8-Q01	complete canonical invariant with certified map family
STAB-04	two-dimensional connected stabilizer	physically incompatible	SYM8-Q02	failed required certificates: cert_charge; cert_response
RESP-02	stabilizer-adapted canonically normalized neutral basis	lawful redescription	SYM8-Q04	complete canonical invariant with certified map family

Audit ID	Challenge	Disposition	Decisive result	Audit basis
RESP-04	second neutral null direction	physically incompatible	SYM8-Q03	failed required certificates: cert_charge; cert_response; cert_spectrum
CHG-03	common sign reversal of Q and electromagnetic unit	lawful redescription	SYM8-Q05	complete canonical invariant with certified map family
ACT-04	all residual electromagnetic transformations declared skin	physically incompatible	SYM8-Q06	failed required certificates: cert_boundary
SPEC-04	gauge-dependent real mass parameter treated as invariant resonance pole	physically incompatible	SYM8-Q07	failed required certificates: cert_spectrum
GOLD-04	exact all-energy Goldstone field identity	physically incompatible	SYM8-Q08	failed required certificates: cert_phase; cert_spectrum
HIGGS-04	invertible custodial bidoublet repackaging of the one Higgs carrier	lawful redescription	SYM8-A01;SYM8-A02	complete canonical invariant with certified map family
CUST-03	custodial $SU(2)_V$ identified with residual $U(1)_{em}$	physically incompatible	SYM8-A02	failed required certificates: cert_custodial; cert_phase
PHASE-04	one-doublet and two-doublet models identified as one phase	target update	SYM8-F01;SYM8-Q09	target relation is UPDATE

The table is intentionally selective in the main text; the full fifty-row register appears in the project data. Several examples illustrate the governing discipline.

- A proper-gauge rotation of H_0 is lawful only when the annihilator, response, and phase representative are conjugated with it.
- A rescaling of Q is conventional only when the electromagnetic unit and current pairing are treated consistently.
- A neutral-field rotation is lawful only when the response form, current, and spectral labels are transported by the same transformation.
- A gauge-dependent propagator parameter is not a physical mass certificate; a physical pole or invariant correlator is required.
- The exact custodial limit at $g_1 = 0$ is a controlled target fiber, not the residual electromagnetic group of the physical core target.

15.5 Counterexample and open-burden protocol

A genuine same-target counterexample must specify D_{EW}^{1H} without alteration, provide a complete physical packet, satisfy the imported premises, and show a quotient-level difference not removed by $\sim_{adm}$. It must then identify the first failed implication among SYM8-Q01–SYM8-Q09, SYM8-A01, SYM8-A02, or the integrated theorem SYM8-A04. Such a counterexample is not redirected by label. It reopens the relevant theorem.

An unregistered proposal is handled in the same way. The author must first determine whether it changes target or presentation. If neither, it enters the open-burden ledger until its physical status is proved. This fail-open rule is the correct boundary between a useful finite audit and an exhaustion claim.

Exact nonclaim boundary

SYM8-A03 proves the completeness and internal consistency of the published fifty-row audit. It does not establish semantic candidate completeness, a universal classification result, or the absence of every possible same-target rival. Those are not premises of the scientific normal-form theorem.

16 Integrated electroweak normal-form rigidity**16.1 The complete physical packet**

The preceding results are assembled without adding a new physical premise. Define

$$\begin{aligned} \mathfrak{E}_{EW} = & (D_{EW}^{1H}, [H_0]_{\mathcal{G}_{0,0}}, [u(1)_Q]_{\text{conj}}, C_g, [\mu_{H_0}]_K, \\ & [A_\mu], [Z_\mu], [W_\mu^\pm], \mathcal{Q}_{\text{phys}}, \mathcal{A}_{\text{em}}, \mathcal{P}_{\text{phys}}, \mathcal{T}_{G/L}, \\ & C_H, C_{\text{cust}}, \Sigma_{\text{phase}}). \end{aligned} \quad (25)$$

The brackets on the gauge-field entries denote transported response and pole classes in a declared canonical kinetic presentation. They do not promote a literal field formula into primitive identity.

For any presentation E of the selected target, let $\mathcal{I}_8(E)$ denote the tuple of quotient-stable data appearing in (25): fixation orbit, stabilizer conjugacy class, coupling map, response rank and kernel, charge ratios, action locus, spectral packet, Goldstone translation, carrier status, custodial status, and phase signature.

16.2 Component rigidity

Lemma 16.1 (Fixation component). *Every admitted presentation contains the same nonzero physical fixation class up to proper gauge equivalence.*

Proof. By SYM8-Q01, proper-gauge-related representatives collapse to one quotient class. A zero branch or extension branch changes the target fingerprint. A gauge condition or gauge-variant expectation value has no independent selector authority. $\square$

Lemma 16.2 (Residual and response components). *At D_{EW}^{1H} , the connected annihilator is $\mathbb{R}Q$ up to conjugacy and normalization, and the Higgs-induced response form is positive semidefinite of rank three with field-coordinate kernel*

$$\mathbb{R}Q_g = C_g^{-1}(\mathbb{R}Q).$$

These statements are invariant under kinetic-metric-aware neutral-basis transport.

Proof. The annihilator calculation SYM8-Q02 fixes its dimension and generator line. The response construction SYM8-Q03 fixes its kernel and rank. The neutral normal-form theorem SYM8-Q04 fixes one null and one nonzero neutral direction after canonical normalization. Hypercharge normalization covariance SYM8-F02 preserves the interaction pairing and the relation between generator and field-coordinate spaces. $\square$

Lemma 16.3 (Charge and action components). *Every admitted presentation contains the electromagnetic charge ratios of SYM8-Q05 and the proper-local/boundary locus split of SYM8-Q06, modulo a common unit or sign convention and lawful hypercharge-normalization transport.*

Proof. The Higgs-annihilator line and current projection fix Q . Acting on the inherited field registry fixes all ratios. The normalization transformation preserves $g_1 Y$, the generator line, and the ratio ledger. The residual-action status is not inferred from the group name: proper operations are quotient-trivial, whereas a nontrivial boundary action requires the surface-charge certificate. $\square$

Lemma 16.4 (Spectrum and translation components). *At the explicit target bundle $\mathcal{S}_{\text{pert}}$, every admitted presentation reconstructs the same certificate-conditional perturbative physical response spectrum and the same regime-scoped Goldstone–longitudinal translation.*

Proof. The response nullity and charge labels are local results. The BRST, physical-pole, field-strength, and invariant-correlator existence premises are explicit entries of $\mathcal{S}_{\text{pert}}$. Together they determine the packet of SYM8-Q07. The translation theorem SYM8-Q08 fixes the relation between gauge-fixed angular variables and longitudinal amplitudes, including finite factors and the $O(m_V/E)$ remainder. Gauge-fixing and chart changes preserve the physical certificates under the stated hypotheses. $\square$

Lemma 16.5 (Structural components). *Every admitted presentation preserves the inherited Higgs carrier, keeps the custodial packet distinct from electromagnetism, and carries the same phase signature.*

Proof. The Higgs compatibility theorem SYM8-A01 blocks the field, vev, pole, potential, Goldstone chart, gauge, or bidoublet display from becoming a second identity ground. The custodial noncollapse theorem SYM8-A02 fixes a separate target-indexed global or limit structure. The phase theorem SYM8-Q09 identifies gauge, basis, and chart changes as representative precisely when the invariant physical packet is preserved. $\square$

16.3 Integrated theorem

Theorem 16.6 (One-doublet electroweak normal-form rigidity; SYM8-A04). *Fix D_{EW}^{1H} , including the explicit perturbative spectral premise bundle $\mathcal{S}_{\text{pert}}$, and assume the imported interfaces listed in section 1. Then:*

1. *the quotient fixation class has a unique connected electromagnetic annihilator line up to conjugacy and normalization;*
2. *the Higgs-induced gauge response has rank three and one-dimensional coupling-pullback kernel, yielding one electromagnetic null direction and the $W^\pm, Z$ response directions;*
3. *the inherited electric-charge ratios, the proper-local/boundary action split, the certificate-conditional perturbative spectrum, the Goldstone–longitudinal translation, the Higgs-carrier status, the custodial status, and the phase signature are fixed as stated in SYM8-Q01–SYM8-Q09 and SYM8-A01–SYM8-A02;*
4. *for every lawful presentation change $E \sim_{\text{adm}} E'$,*

$$\mathcal{I}_8(E) = \mathcal{I}_8(E') = \mathcal{I}_8(\mathfrak{E}_{\text{EW}}).$$

Thus ordinary gauge, weak-basis, normalization, neutral-basis, gauge-fixing, Goldstone-chart, and invertible one-doublet repackaging choices do not generate a second physical electroweak occupant.

Proof. Existence follows from the explicit construction: the target and field registries are fixed; the quotient fixation class is nonempty; the annihilator and response form are calculated; the neutral directions and current projection are obtained; the charge ratios and residual loci are determined; the spectral and Goldstone premises are integrated; and the Higgs, custodial, and phase interfaces are fixed.

The first three clauses are exactly the component lemmas above. For the fourth clause, every generator of $\sim_{\text{adm}}$ preserves the target and all dependent interfaces by theorem 3.3. The invariant signature is therefore constant along every finite composition of lawful presentation changes. The resulting differences are changes of representation, not changes of the physical packet. $\square$

16.4 What is fixed and what is not

The theorem fixes:

1. the nonzero fixation quotient class;
2. the connected residual generator line and its point-sector charge action;
3. response rank three and kernel one, with the kernel typed in field-coordinate space and related to Q by C_g ;

4. one electromagnetic null direction and three massive vector-response directions;
5. the point-sector charge ratios and the equality $e = g_2 s_W = g_1 c_W$ in canonical normalization;
6. the proper-local/boundary residual-action split;
7. the target-certified perturbative physical sector packet;
8. the scoped Goldstone–longitudinal amplitude translation;
9. preservation of the Higgs bridge/fixation carrier;
10. noncollapse of custodial and electromagnetic structures;
11. the invariant phase signature.

It does not fix a gauge, Higgs orientation, weak-flavor basis, hypercharge unit, sign convention, neutral-field notation, gauge parameter, Goldstone chart, potential parametrization, or bidoublet display.

16.5 Falsification boundary

The theorem is broken by any one of the following:

- a same-target nonzero fixation construction with a connected annihilator not conjugate to $u(1)_Q$;
- a same-target response form satisfying the fixed kinetic and representation data but with different rank or coupling-pullback kernel;
- a charge action preserving all inherited interactions but with different charge ratios;
- a residual operation with the same boundary certificate but a different quotient action;
- a physical spectrum packet satisfying S_{pert} but incompatible with the response, charge, or gauge-independence conclusions;
- a same-target Higgs, custodial, or phase construction preserving the premises but changing the invariant role packet;
- a claimed presentation transformation that fails to transport its dependent data or fails to preserve a physical certificate.

A target change, unsupported assertion, or alternative notation is not such a falsifier. Conversely, an omitted same-target construction is not excluded by the finite audit; it must be assessed against these theorem burdens.

Endpoint in physics-facing form

At the declared one-doublet electroweak scope, the physical content commonly compressed into “gauge symmetry breaking” is a quotient-stable normal form: a Higgs fixation class; a unique connected electromagnetic stabilizer; a rank-three response with one photon direction; fixed charge ratios; a locus-sensitive electromagnetic action; a target-certified gauge-independent perturbative response spectrum; and a controlled Goldstone translation. The field, vev, gauge, basis, and chart choices are indispensable calculational representations, but they do not multiply the physical construction when all dependent data are transported lawfully.

Exact nonclaim boundary

The theorem is not a candidate-completeness or exhaustion theorem. It proves the rigidity and presentation covariance of the one-doublet electroweak construction under its stated premises. A universal claim that no other same-target construction can exist would require an independent completeness argument not supplied by the finite registry or validator.

17 Adversarial audit and theorem-breaking burdens

17.1 Audit rule

A correctness-relevant criticism must identify a false premise, invalid calculation, failed specialization, non-lawful quotient, unsupported physical certificate, hidden target change, provenance failure, or same-target construction that violates the integrated theorem. A preference for conventional phrasing, an objection to AASC vocabulary, or the existence of a legitimate richer theory is not by itself a break. Conversely, a passing artifact audit does not immunize a scientific claim: the theorem remains answerable to its mathematical and physical proof.

17.2 Principal objections

Table 12: Principal adversarial objections and exact disposition.

Objection	Status	Required resolution
A nonzero Higgs vev proves spontaneous breaking of local gauge symmetry.	Rejected.	The vev is gauge-variant and chart-dependent. Physical content is carried by the quotient fixation class, response form, and gauge-invariant or gauge-independent certificates.
Elitzur’s theorem makes electroweak symmetry breaking meaningless.	Rejected.	It blocks an ordinary local gauge-variant order parameter; it does not remove gauge-invariant phase structure, response, poles, charges, or boundary actions.
The residual generator is inserted by hand.	Rejected.	Q is obtained from the annihilator equation $QH_0 = 0$, with convention-neutral normalization.
The photon direction is literally the same vector as Q .	Type error.	$Q \in \mathfrak{g}_{EW}$, while the null field direction lies in gauge-field coefficient space and equals $C_g^{-1}(\mathbb{R}Q)$.
A different A/Z rotation is equally valid.	Qualified.	It is a lawful basis change only when the response form, current pairing, kinetic metric, and spectral labels are transported together.
All residual electromagnetic transformations are physical.	Rejected.	Proper local transformations remain quotient-trivial. A nontrivial action requires the stated boundary or asymptotic charge certificate.
Gauge-fixed fields are not observables, so no spectrum can be stated.	Overbroad.	The spectrum is stated through BRST cohomology, physical amplitudes, invariant correlators, and admitted gauge-independent pole classes.
Response rank proves the existence of all physical poles.	Rejected.	Pole and invariant-correlator existence are explicit premises of $\mathcal{S}_{\text{pert}}$; rank organizes them but does not create them.
The Goldstone equivalence theorem is an exact all-energy field identity.	False.	The theorem retains finite factors, the high-energy regime, and the $O(m_V/E)$ remainder.

Objection	Status	Required resolution
Custodial $SU(2)_V$ is the residual electromagnetic group.	False.	The two structures have different targets, acting objects, dimensions, quotient loci, and deformation ancestry.
The finite registry proves that no other candidate exists.	Rejected by design.	The registry is a declared audit. Semantic completeness and exhaustion require an independent theorem and remain downstream burdens.
AASC terminology is optional.	Irrelevant to the QFT spine.	The stabilizer, response, charge, spectrum, and amplitude results stand in conventional language; AASC classifies their structural roles afterward.

17.3 Dependency-inversion audit

The following inference patterns are prohibited:

- using the words “skin” or “carrier” to establish a quotient or spectrum fact;
- using the familiar name $U(1)_{\text{em}}$ to bypass the annihilator calculation;
- using a stored audit disposition to prove physical compatibility;
- using response rank to infer spectral existence without S_{pert} ;
- using a gauge-fixed pole or field count as a quotient-level spectrum certificate;
- using a finite registry or an empty counterexample file as a semantic completeness theorem;
- using validator success as a substitute for the proof of SYM8-A04.

17.4 Certification scope

The project certification layer checks only predicates that are explicitly encoded and appropriate to automation:

1. target, theorem, source, import, atlas, and nonclaim registry well-formedness;
2. exact rational charge identities and the separation of generator and field-coordinate spaces;
3. response-matrix rank, null vector, and hypercharge-normalization covariance;
4. theorem IDs, labels, dependencies, and source locators;
5. consistency between machine-readable data and rendered manuscript tables;
6. artifact hashes, clean rebuild, citation resolution, PDF preflight, and visual rendering;
7. hostile corruptions of the encoded algebra, provenance, schema, and publication boundary.

It does not infer whether an unencoded transformation is physically gauge or global, prove that the rival registry is semantically complete, or pronounce an exhaustion theorem.

Hostile-audit result

The isolated hostile-audit suite passed **16/16** declared corruption tests. The suite confirms that the encoded target, algebra, provenance, synchronization, nonclaim, citation, overclaim, and manifest gates fail closed under the registered mutations. It does not simulate every possible physical rival or replace specialist review of the theorem chain.

17.5 Theorem-breaking burden

A decisive same-target objection must identify the first failed implication in the scientific chain:

$$\begin{aligned} \text{quotient fixation} &\implies \text{annihilator} \implies \text{response kernel and rank} \\ &\implies \text{charge and action} \implies \text{certified spectrum} \\ &\implies \text{Goldstone translation} \implies \text{structural integration.} \end{aligned}$$

An objection that merely requests a global-form completion, a multi-Higgs theory, a nonperturbative charged Hilbert space, or a whole-Standard-Model candidate inventory changes the endpoint rather than breaking this chain.

Referee boundary

The electroweak theorem is accepted or rejected on its physical and mathematical argument. Certification failures can block publication by showing that the stated proof, source, or artifact is not faithfully represented. Certification success cannot strengthen the theorem beyond the proof actually supplied.

18 Conclusion and downstream interfaces

18.1 Scientific conclusion

The one-doublet electroweak phase does not require an ontic breaking of local gauge redundancy. Proper local gauge-related Higgs configurations represent one physical fixation class. The selected nonzero branch has a unique connected annihilator line generated by

$$Q = T_3 + \frac{1}{2y_H}Y,$$

and the Higgs kinetic term induces a positive-semidefinite gauge-response form of rank three whose one-dimensional field-coordinate kernel is the coupling pullback of that line. In canonical neutral coordinates, this yields the electromagnetic and Z -response directions; the charged directions are $W^\pm$. Acting with Q on the inherited representations reconstructs the electric-charge ratios, while the residual action separates quotient-trivial proper-local operations from conditionally physical boundary or asymptotic transformations.

The response calculation is kept distinct from spectral existence. At the declared spectral bundle S_{pert} , BRST, field-strength, pole, amplitude, and invariant-correlator certificates organize one massless electromagnetic vector sector, three massive vector-response sectors, and one neutral scalar response channel. Goldstone coordinates remain legitimate computational representatives of the longitudinal sectors through the high-energy equivalence theorem, with finite factors and controlled corrections. They do not create extra physical occupants. The Higgs bridge/fixation carrier is preserved without promoting the field, vev, potential, pole, or chart into a second identity ground, and custodial $SU(2)_V$ remains a separately typed global or limit structure.

These results assemble into the presentation-covariant electroweak normal form $\mathfrak{E}_{\text{EW}}$. Gauge, weak-basis, hypercharge-normalization, neutral-basis, gauge-fixing, Goldstone-chart, and invertible one-doublet repackaging choices alter the display but preserve the invariant packet when their dependent data are transported lawfully.

18.2 Relation to the original SYM-8 plan

The original plan required seven results: local-gauge non-order-parameter status; vacuum stabilizer; Higgs bridge compatibility; Goldstone translation; residual electromagnetic action; custodial separation;

and a phase-versus-presentation rule. All seven are established. The manuscript strengthens that target by adding a coordinate-free response form, an explicit coupling map, a rank/kernel theorem, normalization-covariant charge reconstruction, the point-sector charge ledger, a precise proper-local/boundary action split, and a certificate-conditional perturbative spectrum theorem.

The strengthening remains sector-local. It does not move whole-arc candidate completeness or exhaustion into SYM-8. The accompanying fifty-row rival audit demonstrates that the principal foreseeable alternatives have been treated consistently, but it is not used as a proof that every possible same-target candidate has been generated.

18.3 Exports to SYM-9

The following objects pass forward to the global-form and generalized-symmetry analysis:

1. the connected Lie-algebra generator line $\mathbb{R}Q$;
2. the point-particle charge pairing and charge ratios;
3. the local response kernel and photon direction;
4. the proper-local/boundary distinction;
5. the explicit nonclaim that these data do not determine the faithful global electromagnetic subgroup, central intersections, line operators, monopole sectors, theta periodicities, one-form symmetries, or higher structures.

SYM-9 must therefore preserve the local results while deciding which global forms and extended-operator sectors realize them.

18.4 Exports to SYM-11

SYM-11 may import:

1. the theorem ladder SYM8-F01–SYM8-Q09 and SYM8-A01–SYM8-A04;
2. the complete thirty-field SYM-8 atlas rows;
3. the target, field, charge, source, import, theorem, rival, and nonclaim registries;
4. the declared finite rival audit and its exact non-exhaustion boundary;
5. the presentation-covariant normal-form packet $\mathfrak{G}_{EW}$.

The capstone must independently prove its own candidate-generation completeness, role coverage, support-graph minimality, and same-scope rival disposition across the entire symmetry atlas. No local sector registry may be promoted into that proof merely by aggregation.

18.5 Final statement

The standard electroweak language is retained but typed. The Higgs vev is a useful representative; the fixation orbit and response form carry the quotient-stable content. The phrase “broken gauge group” is translated into a connected annihilator, a rank-three response, and a residual physical charge structure. The photon and Z are not arbitrary names but the null and image directions of the neutral response with the current pairing transported consistently. Goldstone fields remain indispensable in perturbation theory without being counted twice. Custodial symmetry remains physically useful without being confused with electromagnetism. The resulting picture preserves ordinary calculations while removing the category errors that arise when redundancy, residual action, vacuum fixation, spectrum, and structural role are compressed into one phrase.

Final endpoint

SYM-8 establishes fixed-target one-doublet electroweak vacuum, residual-action, charge, and certificate-conditional spectrum rigidity under lawful presentation change. The theorem is grounded

in the quotient, annihilator, response, charge, spectrum, amplitude, Higgs-carrier, custodial, and phase arguments. The project artifacts certify that this argument is scoped, sourced, algebraically encoded, reproducible, and faithfully rendered; they do not replace the argument or elevate the finite rival audit into an exhaustion theorem.

V

Audit appendices

A Notation, conventions, and stable theorem identifiers

A.1 Hypercharge convention

The core manuscript uses

$$D_\mu = \partial_\mu - ig_2 W_\mu^a T_a - ig_1 B_\mu Y, \quad YH = \frac{1}{2}H, \quad Q = T_3 + Y.$$

For an arbitrary nonzero Higgs hypercharge coordinate y_H , the convention-independent generator is

$$Q = T_3 + \frac{1}{2y_H}Y.$$

Under $Y \mapsto \lambda Y$, $y_H \mapsto \lambda y_H$, and $g_1 \mapsto g_1/\lambda$, both Q and $g_1 Y$ are invariant. A common reversal $(Q, e) \mapsto (-Q, -e)$ is a reporting convention. Charge ratios are invariant.

A.2 Core symbols

Table 13: Core notation.

Symbol	Meaning
D_{EW}^{1H}	Frozen one-Higgs-doublet electroweak target.
$\mathcal{G}_{0,\partial}$	Proper local gauge transformations compatible with the admitted boundary data.
$[H_0]_{\mathcal{G}_{0,\partial}}$	Physical nonzero fixation class at the proper local quotient.
$\text{Ann}(H_0)$	Lie-algebra annihilator of the chosen representative.
Q	Electromagnetic generator line, normalized as $T_3 + Y$ in the core convention.
$\mathfrak{a}_{\text{EW}}$	Real gauge-field coefficient space, distinguished from the generator algebra after coupling insertion.
C_g	Invertible coupling map $\mathfrak{a}_{\text{EW}} \rightarrow \mathfrak{g}_{\text{EW}}$.
$Q_g = C_g^{-1}(Q)$	Field-coordinate representative of the electromagnetic response-kernel line.
μ_{H_0}	Positive-semidefinite gauge-response quadratic form on $\mathfrak{a}_{\text{EW}}$.
s_W, c_W	Canonical weak-angle coordinates, $s_W = g_1/\sqrt{g_1^2 + g_2^2}$, $c_W = g_2/\sqrt{g_1^2 + g_2^2}$.
$A_\mu, Z_\mu, W_\mu^\pm$	Canonical representatives of the electromagnetic, massive neutral, and charged response directions.
Q_{phys}	Physical point-sector charge-ratio ledger.

Symbol	Meaning
$\mathcal{A}_{\text{em}}$	Locus-indexed electromagnetic action packet.
$\mathcal{P}_{\text{phys}}$	BRST/pole/amplitude/correlator-certified perturbative physical spectrum packet.
$\mathcal{T}_{G/L}$	Goldstone/longitudinal translation packet including finite factors and remainder.
C_H	Imported Higgs bridge/fixation carrier $H_{\text{br}/\text{fix}}$.
C_{cust}	Target-indexed custodial packet imported from SYM-6.
Σ_{phase}	Invariant SYM-8 phase signature.
$\mathfrak{C}_{\text{EW}}$	Complete canonical electroweak normal-form packet.
$\mathcal{C}_8(D_{\text{EW}}^{1H})_{\text{audit}}$	Declared finite rival registry organized by the ten SYM-8 audit loci.
$\sim_{\text{adm}}$	Lawful presentation equivalence generated by target-preserving gauge, basis, normalization, gauge-fixing, and repackaging transformations.

A.3 Stable theorem ladder

Table 14: Stable SYM-8 theorem ladder.

ID	Result	Proof status	Dependencies
SYM8-F01	Fixed target and target-update separation	local	target registry; SYM-0
SYM8-F02	Hypercharge normalization covariance	local	coupling pairing
SYM8-Q01	Local-gauge non-order-parameter theorem	local+imported	SYM-1; Elitzur; F01
SYM8-Q02	Connected stabilizer uniqueness	local	one-doublet representation; Q01
SYM8-Q03	Gauge-response rank and kernel	local	Higgs kinetic term; Q02
SYM8-Q04	Neutral A/Z normal form	local	Q03; kinetic metric
SYM8-Q05	Residual charge action and ratio rigidity	local	Q02; Q04; F02; representation
SYM8-Q06	Residual electromagnetic locus split	local+imported	SYM-1 boundary grammar; Q05
SYM8-Q07	Certificate-conditional perturbative physical spectrum reconstruction	local conditional on target spectral bundle	Q03/Q04; BRST; admitted pole/correlator bundle $\mathcal{S}_{\text{pert}}$
SYM8-Q08	Goldstone/longitudinal translation	conditional	Q07; equivalence theorem
SYM8-A01	Higgs bridge compatibility and no promotion	composed	Higgs endpoint; Q01-Q08
SYM8-A02	Electromagnetic/custodial noncollapse	composed	SYM-6; Q02/Q05
SYM8-Q09	Phase-versus-presentation rule	local	Q01-Q08; presentation groupoid
SYM8-A03	Declared finite rival audit	finite audit proposition; explicitly non-exhaustive	target registry; fifty-row candidate audit; theorem and source ledgers
SYM8-A04	One-doublet electroweak normal-form rigidity	integrated mathematical and physical theorem	F01-F02; Q01-Q09; A01-A02; presentation-covariance lemma

The stable IDs are normative; typeset theorem numbers may change when editorial sections are

moved. SYM8-F01 and SYM8-F02 are foundation locks. SYM8-Q01–SYM8-Q09 are framework-neutral or QFT-facing results. SYM8-A01–SYM8-A04 are structural integration, declared-audit, and normal-form results.

A.4 Target identifiers

Table 15: Frozen core target and controlled neighboring fibers.

Target ID	Status	Distinguishing datum	Use
D8-CORE-1H	core	One doublet, nonzero branch, nonzero g_1 and g_2 , perturbative spectral certificate bundle	main theorem target
D8-BULK	fiber	proper compact support or identity boundary value	proper-local quotient theorem
D8-BOUNDARY	fiber	local plus admitted asymptotic transformation class	physical electromagnetic charge action
D8-CUST-GP0	controlled limit	custodial target-local	SYM-6 custodial import
D8-CUST-NUD	target update	custodial target-local	Dirac-neutrino custodial branch
D8-V0	target fiber	one doublet	not a same-scope rival
D8-GLOBAL	successor target	faithful quotient, line operators, monopoles and defects specified	SYM-9 handoff
D8-EXTH	target update	declared by extension	outside the core one-doublet target
D8-NONPERT	successor target	must be fixed for complete nonperturbative statement	outside Q07 and the perturbative normal-form theorem

No target identifier is inferred from prose context. In particular, the minimal core excludes ν_R ; the Dirac-neutrino custodial target is separately registered. The global-form target is a successor target, not an unspecified completion of the core.

B Canonical calculations and algebraic certificates

B.1 Doublet generators and annihilator

Use

$$T_1 = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad T_2 = \frac{1}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad T_3 = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad Y = y_H I_2,$$

and

$$H_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}.$$

For $X = a_1 T_1 + a_2 T_2 + a_3 T_3 + b Y$,

$$X H_0 = \frac{v}{2\sqrt{2}} \begin{pmatrix} a_1 - i a_2 \\ -a_3 + 2b y_H \end{pmatrix}.$$

Because $v \neq 0$, annihilation requires $a_1 = a_2 = 0$ and $a_3 = 2b y_H$. Thus

$$\text{Ann}(H_0) = \mathbb{R} \left(T_3 + \frac{1}{2y_H} Y \right).$$

For $y_H = 1/2$, $Q = T_3 + Y$.

B.2 Response form and stabilizer correspondence

Let $\mathfrak{a}_{EW}$ be the gauge-field coefficient space and define

$$C_g(w^a T_a + bY) = g_2 w^a T_a + g_1 bY.$$

For $u, w \in \mathfrak{a}_{EW}$, set

$$\mu_{H_0}(u, w) = 2 \operatorname{Re} \langle \rho_H(C_g u) H_0, \rho_H(C_g w) H_0 \rangle.$$

This form is positive semidefinite and

$$\ker \mu_{H_0} = C_g^{-1}(\operatorname{Ann}(H_0)) = \mathbb{R}Q_g, \quad Q_g = C_g^{-1}(Q).$$

With nonzero g_1, g_2 , C_g is invertible, so the response kernel is one-dimensional and maps isomorphically onto the generator-space stabilizer line. The full response matrix in (W^1, W^2, W^3, B) is

$$M_{EW}^2 = \frac{v^2}{4} \begin{pmatrix} g_2^2 & 0 & 0 & 0 \\ 0 & g_2^2 & 0 & 0 \\ 0 & 0 & g_2^2 & -g_1 g_2 \\ 0 & 0 & -g_1 g_2 & g_1^2 \end{pmatrix}.$$

Its generic rank is three. The neutral block has determinant zero and trace $v^2(g_1^2 + g_2^2)/4$.

B.3 Neutral eigenvectors

The vectors

$$v_A = \begin{pmatrix} g_1 \\ g_2 \end{pmatrix}, \quad v_Z = \begin{pmatrix} g_2 \\ -g_1 \end{pmatrix}$$

satisfy

$$M_0^2 v_A = 0, \quad M_0^2 v_Z = \frac{v^2}{4}(g_1^2 + g_2^2)v_Z.$$

Normalize by

$$s_W = \frac{g_1}{\sqrt{g_1^2 + g_2^2}}, \quad c_W = \frac{g_2}{\sqrt{g_1^2 + g_2^2}}.$$

Then

$$\begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix} = \begin{pmatrix} s_W & c_W \\ c_W & -s_W \end{pmatrix} \begin{pmatrix} W_\mu^3 \\ B_\mu \end{pmatrix}.$$

Equivalent sign conventions are admitted when the current coupling is transported consistently.

B.4 Current projection

Substitution of

$$W^3 = s_W A + c_W Z, \quad B = c_W A - s_W Z$$

into $g_2 T_3 W^3 + g_1 Y B$ gives

$$\begin{aligned} & (g_2 s_W T_3 + g_1 c_W Y) A + (g_2 c_W T_3 - g_1 s_W Y) Z \\ &= e(T_3 + Y) A + \frac{g_2}{c_W} (T_3 - s_W^2 Q) Z, \end{aligned}$$

where

$$e = g_2 s_W = g_1 c_W.$$

This proves that the response null direction and the physical charge generator are linked components of one packet through C_g and the current pairing; they are not literally the same vector in one untyped space and are not independently selected.

B.5 Charge table

The generated field registry yields:

Table 16: Electromagnetic charge ledger in the convention $Q=T_3+Y$.

Field/component	T3	Y	Q	Certificate
u_L	1/2	1/6	2/3	$Q=T_3+Y$
d_L	-1/2	1/6	-1/3	$Q=T_3+Y$
u_R	0	2/3	2/3	$Q=T_3+Y$
d_R	0	-1/3	-1/3	$Q=T_3+Y$
nu_L	1/2	-1/2	0	$Q=T_3+Y$
e_L	-1/2	-1/2	-1	$Q=T_3+Y$
e_R	0	-1	-1	$Q=T_3+Y$
H_plus	1/2	1/2	1	$Q=T_3+Y$
H_zero	-1/2	1/2	0	$Q=T_3+Y$
W_plus	<i>adjoint + 1</i>	0	1	$[Q, T_+] = +T_+$
W_minus	<i>adjoint - 1</i>	0	-1	$[Q, T_-] = -T_-$
A	<i>neutral</i>	<i>neutral</i>	0	kernel generator field
Z	<i>neutral</i>	<i>neutral</i>	0	neutral image field

The generated table is the core electromagnetic charge ledger; it contains no ν_R row. The controlled Dirac-neutrino target remains separately registered in the field and target registries. The charged vector generators $T_{\pm} = T_1 \pm iT_2$ satisfy

$$[Q, T_{\pm}] = [T_3, T_{\pm}] = \pm T_{\pm},$$

so $Q(W^{\pm}) = \pm 1$, while the neutral gauge directions have zero adjoint electric charge.

B.6 Degree-count certificate

Before fixation, the bosonic coordinate count is four real scalar coordinates plus four massless gauge fields with two physical polarizations each:

$$4 + 4 \times 2 = 12.$$

After fixation, the perturbative physical count is three massive vector sectors with three polarizations, one massless vector sector with two polarizations, and one radial scalar channel:

$$3 \times 3 + 1 \times 2 + 1 = 12.$$

This is a consistency certificate, not an independent proof of the spectrum. BRST cohomology and pole/amplitude certificates determine physical status.

B.7 Symbolic validation results

Encoded validation result

The publication validator passed **71/71** declared checks. The checks cover registry identity, target and source completeness, theorem-label and citation synchronization, exact charge arithmetic, symbolic response identities, generator/field-space typing, normalization covariance, generated-table identity, and the publication claim boundary. This result certifies the encoded artifact layer only; the scientific theorems retain the proof authority stated in the manuscript.

The JSON report records each individual check, its result, the frozen counts, and the approximate source word count. The publication manifest records artifact hashes after the final PDF has passed visual preflight.

C Canonical atlas and declared rival audit

C.1 Thirty-field atlas

The canonical atlas contains sixteen integrated SYM-8 records. Every row implements the full thirty-field ARC schema: stable identity and aliases; exact domain and background; candidate kind; acting and acted-upon objects; action mode; invariant object; physical quotient and quotient action; current, constraint, and charge data; independent classical, quantum, anomaly, vacuum, breaking, scale, operator, and global-form fields; AASC role and role proof; carrier support; evidence status; source certificate; rival status; nonclaims; paper ownership; and change history. A readable projection appears in [table 8](#); it is not a replacement for the CSV/JSON authority.

C.2 Declared rival audit

The rival audit contains fifty concrete schemas distributed across the ten loci of [table 3](#). The authoritative file `data/candidate_audit.csv` records the target relation, audit disposition, decisive test, invariant key where applicable, and source or theorem authority. The dispositions are evidence-backed editorial conclusions about those rows; they are not treated as machine-generated physical truths.

The disposition totals and representative rows appear in [tables 10](#) and [11](#). The full row-level file remains the publication authority.

The row set is deliberately finite. A proposal absent from the registry is not thereby rejected. It enters the same target, quotient, certificate, and theorem analysis used for the published rows. This preserves the audit's usefulness without converting it into a candidate-completeness theorem.

C.3 Rival classes

The broader prose rival ledger groups related rows and states the decisive physical result.

Table 17: Scope-preserving rival ledger.

ID	Rival	Audit disposition	Decisive result
R01	Different gauge orientation of H0	same lawful orbit	SYM8-Q01
R02	Gauge condition selects residual subgroup	ineligible	SYM8-Q01
R03	Two connected residual generators annihilate H0	ineligible	SYM8-Q02
R04	Alternative Q with same charge ratios	same lawful orbit	SYM8-F02;SYM8-Q05
R05	Alternative Q with different charge ratios	ineligible	SYM8-Q05

ID	Rival	Audit disposition	Decisive result
R06	Second massless neutral vector	ineligible	SYM8-Q03;SYM8-Q04
R07	Different A/Z rotation with same invariant packet	same lawful orbit	SYM8-Q04
R08	Different rotation with currents not transported	jointly incompatible	SYM8-Q04
R09	Local U(1) _{em} operation declared physical everywhere	ineligible	SYM8-Q06
R10	All electromagnetic transformations declared skin	ineligible	SYM8-Q06
R11	Goldstones counted in addition to longitudinal vectors	ineligible	SYM8-Q07;SYM8-Q08
R12	Goldstone equivalence asserted exactly at all energies	ineligible	SYM8-Q08
R13	Higgs vev or pole declared primitive identity	ineligible	SYM8-A01
R14	Bidoublet declared a second Higgs carrier	ineligible under invertible-repackaging and carrier-cardinality certificate	SYM8-A01;SYM8-A02
R15	Custodial SU(2) _V identified with U(1) _{em}	ineligible	SYM8-A02
R16	$\rho=1$ used as biconditional definition of custodial symmetry	ineligible	SYM8-A02
R17	Extra scalar produces a second Higgs at same scope	target update	SYM8-F01;SYM8-A01
R18	$v=0$, $g_1=0$, or $g_2=0$ offered as same-scope rival	target fiber	SYM8-F01
R19	Faithful global subgroup differs while local algebra agrees	target update	SYM8-F01;SYM8-Q02
R20	Line or monopole sector alters residual structure	target update	SYM8-F01
R21	Nonperturbative spectrum invalidates the perturbative packet	target update	SYM8-F01;SYM8-Q07
R22	Same target and all certificates, second non-equivalent packet	open same-target counterexample protocol; not discharged by finite audit	SYM8-A04; identify the first failed load-bearing theorem or a new physical invariant

C.4 Authority and synchronization

The authoritative research objects are the target, field, charge, theorem, source, import, atlas, nonclaim, rival, presentation-map, and candidate-audit files. Rendered tables are deterministic projections. The validator checks that the projections agree with the machine-readable sources, that all stable identifiers are unique, and that the manuscript does not claim semantic completeness or exhaustion from the finite registry.

A prose statement that disagrees with a controlled row is a publication defect and must be reconciled. Neither the data file nor the prose acquires scientific authority merely by prevailing in that reconciliation; the governing authority is the cited theorem, calculation, or explicit conditional premise.

D Source, dependency, and proof-provenance ledger

D.1 Direct arc dependencies

Table 18: Direct SYM dependencies and their precise use.

Source	Imported interface	Use and noncircularity boundary
SYM-0 [1]	target-update separation; lawful presentation quotient; typed rival disposition	Supplies classification grammar only after QFT residues are constructed. Does not supply the electroweak stabilizer or response.
SYM-1 [2]	proper-local quotient; architecture/operation split; boundary exception; gauge-fixing no-authority discipline	Specialized to the electroweak fixation and residual electromagnetic loci. Does not identify Q .
SYM-3 [3]	gauge anomaly as inconsistency; anomaly-field transport	Fixes anomaly-neutral target status. No anomaly calculation is used to choose the vacuum.
SYM-5 [4]	weak-basis quotient; fixed-tensor stabilizers; invariant extraction	Prevents a Yukawa basis or rephasing from selecting the vacuum, charge action, or residual structure.
SYM-6 [5]	custodial C01–C06 packet: bidoublet, nested targets, diagonal stabilizer, deformation ancestry, diagnostics	Imported as a separate target-indexed symmetry packet; not used to derive electromagnetism.
SYM-7 [6]	fixed minimal/Dirac-neutrino target discipline; operator-order and remnant boundary	Prevents silent ν_R or higher-operator import into the core target.

D.2 Structural support dependencies

Table 19: Additional AASC and Standard Model support.

Source	Imported result	SYM-8 use
Gauge–Vacuum Compatibility [13]	gauge-vacuum compatibility; admissible vacuum-mediated response reduction; exhaustion of failure routes	Blocks gauge fixing or representative choice from manufacturing a residual physical sector.
Vacuum-Compatible Residual Sector Normal Form [14]	residual-sector normal form and import exhaustion	Supplies residual syntax and failure taxonomy without pre-identifying electromagnetism.
Abelian Charge-Ratio Rigidity [15]	charge-ratio standing and normalization-skin discipline	Supports the distinction between physical ratios and arbitrary Abelian coordinate normalization.
Standalone Standard Model Structure [16]	electroweak representation, one-doublet Higgs, charge, anomaly, vacuum, and parameter-status structural tuple	Supplies the field and interaction registry on which local calculations are performed.

Source	Imported result	SYM-8 use
Realized Physical Interior [17]	field, particle, gauge, mass-response, vacuum, and quotient roles	Supports physical quotient and spectrum-role typing.
Higgs Structural Endpoint Arc [7, 8, 9, 10, 11, 12]	bridge/fixation carrier; identity occupation; no primitive scale; vacuum route; extension classifier; integrated endpoint	Fixes Higgs provenance and blocks primitive promotion. Does not substitute for the electroweak QFT calculations.

D.3 Primary QFT certificates

Table 20: External source certificates.

Source	Certificate	Use and limit
Elitzur [18]	local gauge-variant order parameter obstruction	Supports the non-order-parameter interpretation; the manuscript also uses its own quotient theorem.
Becchi–Rouet–Stora and Kugo–Ojima [19, 20]	BRST cohomological physical-state structure	Used for physical-state and double-counting discipline, not as an ordinary global particle symmetry.
Barnich–Brandt–Henneaux [21]	local BRST cohomology	Supports local observable/cohomology distinctions.
Nielsen; Stuart; Sirlin; Gambino–Grassi [22, 30, 31, 23]	gauge-parameter identities and gauge-independent complex-pole language	Used for perturbative physical pole transport after the target spectral bundle is declared; not a universal nonperturbative spectrum theorem.
Fröhlich–Morchio–Strocchi [27, 28]	gauge-invariant composite-channel reconstruction	Used as a controlled example and boundary; no exact universal field equality is asserted.
Gounaris–Kögerler–Neufeld; Yao–Yuan; Bagger–Schmidt; Veltman [24, 25, 26, 32]	longitudinal–Goldstone amplitude equivalence with loop normalization and high-energy regime	Imported with finite factors and remainder.
Fradkin–Shenker [29]	analytic connection in lattice gauge–Higgs systems under specified conditions	Restricts phase overstatement; not used to deny the selected perturbative packet.
Sikivie et al. [33]	custodial symmetry in electroweak models	External conventional-facing support for the target-indexed custodial packet.

D.4 Publication provenance

The source ledger records an immutable source ID, title, source type, exact theorem/page or bibliographic locator, preferred-version/file identifier, SHA-256 hash where a local source file is present, verification date, and publication status. The import ledger separately records the exact result imported, its

target interface, the retained nonclaim boundary, and the local theorem that uses it. The following table is a readable projection; the full hashes and identifiers remain in `data/source_ledger.csv` and `data/import_registry.csv`.

Table 21: Publication source-provenance projection.

Source ID	Source	Exact locator	Imported use
SRC-ARC-PLAN	Standard Model Symmetry Arc Plan	pp. 28-29 (SYM-8 plan); pp. 32-36 (governance); pp. 41-44 (30-field atlas); pp. 48-50 (hostile/-capstone)	target, governance, atlas, and successor boundaries
SRC-SYM0	The Symmetry Status Calculus	SYM0-T2 p.91; SYM0-T9, T10, T13 and Theorem 10.1 p.92	F01; presentation covariance; A03-A04
SRC-SYM1	Gauge Architecture and Gauge Skin in the Standard Model	Theorem 2.2 pp.10-11; 10.2 p.29; 11.1 p.32; 12.1 p.34; 14.2-15.1 pp.41-42	Q01; Q06; presentation and compatibility rules
SRC-SYM3	Anomalies as Structural Obstructions	gauge-consistency and mixed-anomaly ledgers; gauge-anomaly status table in source theorem registry	target/anomaly certificate
SRC-SYM5	Flavor Symmetry, Basis Skin, Spurion, and Invariant Normal Form	basis-quotient theorem pp.25-26 and theorem registry p.52; fixed-tensor stabilizer pp.25-26	presentation maps; Q05/Q09
SRC-SYM6	Chiral, Custodial, and Infrared Symmetries	Theorems 14.1-14.2 p.33; 15.1 p.34; 15.2 p.35; 16.1-16.2 p.37	A02
SRC-GAUGE-VAC	Gauge-Vacuum Compatibility	Theorems 10.1 p.16; 12.1 p.18; 18.1 p.23	Q01/Q03
SRC-RESIDUAL-NF	Vacuum-Compatible Residual Sector Normal Form	Theorem 15.1 p.16; Theorem 16.1 p.18	Q02/Q06
SRC-SR4	Abelian Charge-Ratio Rigidity in the Admissible Gauge Arena	Theorem 24.1 p.20	F02/Q05
SRC-SR7	Standard-Model Realization Rigidity in the Admissible Gauge Arena	Corollary 15.2 p.14; Theorem 19.1 p.17	non-circularity boundary
SRC-STANDALONE-SM	The Standalone Standard Model Structure from First Principles	Theorem 9.7 p.36; representation tables pp.32-33	target and Q05
SRC-PHYS-INTERIOR	The Realized Physical Interior	Theorems 13.1-17.1 pp.17-19; Proposition 27.2 p.28	Q07 and physical-status translation
SRC-R10	Vacuum as Minimal Concrete Persistence Regime	vacuum baseline and non-null carrier ledger, full 12-page source	target/vacuum interpretation
SRC-ET-NOTE	Equivalence Completion for Longitudinal Gauge Modes and Goldstone Degrees	full 2-page note; used only as scoped support	Q08 supporting amplitude translation

Source ID	Source	Exact locator	Imported use
SRC-EWQED-NOTE	Equivalence Completion Between Electroweak and Electromagnetic Coupling Bookkeeping	full 2-page note; used only for EM projection bookkeeping	Q04-Q05 coupling-projection bookkeeping
SRC-H1	The Higgs Bridge Carrier: Kernel Consequences for Electroweak Mass-Response	bridge-carrier necessity theorem; primitive representative collapse; June 30 preferred copy	A01
SRC-H2	Endpoint on Higgs Structural Identity: Bridge-Carrier Occupation and No Fifth Identity Ground	Theorem 7.3 no fifth identity ground; Theorem 8.1 Higgs Structural Identity	A01
SRC-H3	No Primitive Higgs Scale: Carrier-Indexed Scale Standing and Hierarchy Orientation	Theorems 15.3-15.4	A01 nonclaims
SRC-H4	Higgs Vacuum Route Status: Metastability, Lower Branches, and Standing Closure	Higgs Vacuum Endpoint Theorem; RouteAvail does not imply StandingClosed; DOI 10.5281/zenodo.20996895	A01; target discipline
SRC-H5	The Higgs Extension Classifier: Extra Scalars and Same-Scope Carrier Preservation	Theorem 15.2; Theorem 16.2; Corollary 16.3; DOI 10.5281/zenodo.20997077	F01/A01
SRC-H6	The Higgs Domain Endpoint: Carrier, Identity, Scale, Vacuum, Extension, and Projection	Theorem 8.3; Theorem 9.1; DOI 10.5281/zenodo.20997159	A01
SRC-ELITZUR	Elitzur, Impossibility of Spontaneously Breaking Local Symmetries	Phys. Rev. D 12 (1975) 3978-3982	Q01
SRC-BRST	Becchi-Rouet-Stora; Kugo-Ojima; Henneaux-Teitelboim	BRS 1976; Kugo-Ojima 1979; Quantization of Gauge Systems 1992	Q07/Q08
SRC-NIELSEN	Nielsen identity and gauge-independent pole literature	Nielsen 1975; Gambino-Grassi 2000	Q07
SRC-ET	Longitudinal-Goldstone equivalence-theorem literature	Cornwall-Levin-Tiktopoulos; Chanowitz-Gaillard; Bagger-Schmidt and later refinements	Q08
SRC-FMS	Froehlich-Morchio-Strocchi gauge-invariant operator construction	FMS 1980 and 1981	Q07

D.5 Proof provenance rule

The machine-readable theorem ledger distinguishes local proofs, imported-and-specialized results, local theorems conditional on an explicit target bundle, integrated physical/structural theorems, and finite audit propositions. An imported theorem is not relabeled as locally proved. A local calculation may depend on an imported target registry without depending on the conclusion it proves. A conditional

spectrum theorem names its dynamical premises. The finite rival audit is recorded as an audit result and is not promoted into a candidate-completeness or exhaustion theorem. Source chronology or repeated wording never substitutes for the supersession and hash authority.

E Exact nonclaims and standard-language translation

E.1 Exact nonclaim register

SYM-8 does not establish:

1. numerical values of v , g_1 , g_2 , e , θ_W , masses, widths, self-couplings, or vacuum lifetimes;
2. uniqueness of the Standard Model under arbitrary assumptions;
3. the faithful global subgroup $U(1)_{\text{em}}$ inside every admissible global form;
4. a line-operator, monopole, theta-periodicity, higher-form, or noninvertible-symmetry completion;
5. an exact nonperturbative construction of infrared charged states and dressings;
6. a universal equality of elementary gauge-fixed and local gauge-invariant composite correlators;
7. a universal electroweak lattice or finite-temperature phase diagram;
8. exact custodial symmetry of the observed full Standard Model;
9. multi-Higgs, composite-Higgs, strong-electroweak, SMEFT, or BSM classification;
10. a derivation of Higgs identity from the response matrix alone;
11. that gauge-fixed fields, vevs, potentials, ghosts, or Goldstone variables are useless or unreal;
12. that every residual gauge transformation acts physically, or that every residual operation is skin;
13. semantic completeness of the finite rival registry or the complete Standard Model symmetry exhaustion reserved for SYM-11.

E.2 Standard phrase / preserved meaning / blocked promotion

Table 22: Translation of standard electroweak language.

Standard phrase	Preserved physical/calculational meaning	Blocked promotion
"The Higgs gets a vev."	Choose a useful gauge-fixed representative of a nonzero fixation branch; v sets the response scale relative to calibrated inputs.	The literal gauge-variant expectation value is not an ordinary physical order parameter or primitive vacuum selector.
" $SU(2)_L \times U(1)_Y$ breaks to $U(1)_{\text{em}}$."	The nonzero fixation has a one-dimensional connected annihilator; the response form has one null direction; charge and spectrum organize electromagnetically.	Local gauge redundancy is not an ordinary global symmetry with gauge-related physical vacua; faithful global form is not fixed by this shorthand.
"Three generators are broken."	Three gauge directions lie in the nonzero response image and supply massive-vector response sectors.	The phrase does not create three physical broken charges or an ordinary gauge-variant order parameter.

Standard phrase	Preserved physical/calculational meaning	Blocked promotion
“The Goldstones are eaten.”	Angular scalar coordinates reorganize with massless vector coordinates into longitudinal polarizations of massive vectors; high-energy amplitudes admit Goldstone representatives.	No independently standing particles are literally consumed; Goldstones are not counted in addition to longitudinal vectors.
“The photon is the unbroken gauge boson.”	The unique neutral response null direction couples to Q , remains massless in the target, and supports electromagnetic curvature/charge architecture.	The photon is not mere leftover notation, and not every local gauge operation becomes a physical global action.
“The weak mixing angle rotates W^3 and B .”	In a canonical kinetic basis, s_W, c_W parameterize the orthonormal null/image basis and current projection.	A field-only rotation or numerical-angle prediction is not licensed.
“The Higgs boson is the radial field.”	The radial coordinate is the leading perturbative representative of a neutral scalar gauge-invariant/pole channel.	The coordinate or pole does not replace the inherited Higgs carrier/identity.
“Custodial symmetry explains $\rho \simeq 1$.”	A target-indexed global/limit symmetry organizes response relations and diagnostic protection.	$\rho = 1$ or $T = 0$ is not a biconditional definition and custodial $SU(2)_V$ is not electromagnetism.
“Different gauges describe the same physics.”	Lawful gauge-fixing changes preserve BRST cohomology, physical amplitudes, admitted poles, charge, and response invariants.	An arbitrary change that alters a physical certificate is not excused as gauge choice.

E.3 Terminology discipline

The manuscript may use “electroweak symmetry breaking” as conventional shorthand when the surrounding text identifies the quotient-stable content. In theorem statements, “fixation,” “annihilator,” “response image/kernel,” “residual charge action,” and “physical spectrum” are preferred because they state which object is doing the work.

F Reproducibility, validation, and certification boundary

F.1 Project structure

The publication project is organized as follows:

sections/	the theorem-bearing manuscript sections;
appendices/	notation, calculations, atlas, provenance, nonclaim, reproducibility, and hostile-audit appendices;
data/	target, field, charge, theorem, source, import, atlas, presentation-map, rival, nonclaim, and candidate-audit records;
generated/	deterministic LaTeX projections of the machine-readable records;

scripts/	table generation, validation, and isolated hostile-audit checks;
audits/	machine-readable validation, hostile-audit, PDF-preflight, and render reports;
manifest.sha256	artifact-integrity manifest for the publication package.

F.2 Build sequence

The reproducible build is:

1. generate manuscript tables from the controlled CSV and JSON records;
2. validate registries, theorem identities, sources, imports, citations, encoded algebra, audit boundaries, and generated-table identity;
3. compile the manuscript with `latexmk` and `biber`;
4. preflight the PDF, verify embedded fonts, and render all pages for visual inspection;
5. write the final validation and PDF reports;
6. generate the SHA-256 manifest and archive the project.

The command-line entry points are:

```
make tables
make validate
make pdf
make hostile
make release
```

F.3 Encoded mathematical checks

The validator checks the algebra that is explicitly represented in the project data or source:

1. exact rational identities $Q = T_3 + Y$ for the field and charge ledgers;
2. uniqueness of record, theorem, source, and target identifiers;
3. response-matrix determinant, rank, and null-vector identities for symbolic nonzero g_1, g_2 ;
4. the typed relation between the generator line $\mathbb{R}Q$ and the field-coordinate kernel $C_g^{-1}(\mathbb{R}Q)$;
5. hypercharge-normalization covariance of $g_1 Y$, Q , and the charge-ratio ledger;
6. theorem-ID and LaTeX-label agreement;
7. citation-key resolution against the bibliography;
8. row-level provenance, retained-boundary, and nonclaim completeness;
9. identity between the machine-readable records and their generated LaTeX projections.

F.4 Certification doctrine

The validator does not decide unencoded physical questions. In particular, it does not:

- infer whether an arbitrary transformation is gauge, global, or physical;
- establish the validity of an imported QFT theorem;
- prove that the fifty-row rival registry is semantically complete;
- infer a universal equivalence class from matching labels;
- pronounce a candidate-exhaustion theorem.

Those burdens remain with the cited source or the mathematical and physical proof in the manuscript. The artifact layer verifies that the stated argument, data, and publication package agree.

F.5 Hostile-audit contract

The isolated hostile-audit suite corrupts only predicates that the certification layer claims to enforce. It tests, among other things, altered charge entries, duplicated stable IDs, missing source locators, broken theorem labels, stale generated tables, removed target fields, a false response null vector, an untransported hypercharge convention, an omitted nonclaim, and insertion of forbidden exhaustion language. Each corruption must be rejected at the corresponding gate.

Encoded validation result

The publication validator passed **71/71** declared checks. The checks cover registry identity, target and source completeness, theorem-label and citation synchronization, exact charge arithmetic, symbolic response identities, generator / field-space typing, normalization covariance, generated-table identity, and the publication claim boundary. This result certifies the encoded artifact layer only; the scientific theorems retain the proof authority stated in the manuscript.

F.6 PDF verification

The final PDF is rendered page by page and checked for clipped text, overlaps, broken glyphs, malformed tables, undefined references, and missing fonts. The PDF report records page count, font embedding, metadata, preflight warnings, and render status. Visual verification is a publication-integrity check; it does not bear scientific proof authority.

Reproducibility statement

A clean project extraction regenerates the tables, passes the encoded validation and hostile-audit suite, rebuilds the manuscript, and reproduces the publication PDF within the declared toolchain. The manifest verifies artifact identity after the final build.

G Hostile-audit cases

G.1 Purpose

The hostile-audit suite tests the publication package at its declared certification boundary. It is not a synthetic physics engine. Each case corrupts an encoded datum or artifact relation and asks whether the relevant validator gate rejects the result.

Table 23: Encoded hostile-audit cases and expected gates.

ID	Mutation class	Corruption	Expected gate
H01	duplicate-theorem-id	Duplicate a stable theorem identifier	theorem registry uniqueness
H02	missing-theorem-label	Remove the LaTeX label for SYM8-A04	theorem/source synchronization
H03	charge-mismatch	Alter one Q value in the charge ledger	exact charge algebra
H04	field-charge-mismatch	Alter a field T3 or Y entry without updating Q	field/charge join
H05	response-null-failure	Replace the symbolic photon null vector by a non-null vector	response algebra
H06	generator-kernel-type-collapse	Identify Q coordinates with the field-coordinate kernel without C_g	typed coupling-map check
H07	normalization-failure	Rescale Y without contragredient g_1 transport	normalization covariance

ID	Mutation class	Corruption	Expected gate
H08	missing-source-locator	Blank a load-bearing source locator	provenance completeness
H09	missing-retained-boundary	Remove an import nonclaim boundary	import boundary completeness
H10	duplicate-atlas-row	Duplicate an atlas record ID	atlas identity
H11	stale-generated-table	Modify a generated table without changing source data	table/source identity
H12	missing-nonclaim	Delete the finite-audit non-exhaustion nonclaim	claim boundary
H13	exhaustion-language-insertion	Insert a claim that the validator proves semantic exhaustion	forbidden-overclaim scan
H14	target-field-removed	Remove the spectral or boundary field from the core target	target completeness
H15	citation-key-missing	Insert an unresolved bibliography key	citation resolution
H16	manifest-corruption	Alter a manifested artifact after publication packaging	artifact identity

G.2 Interpretation

The cases fall into five groups:

1. **Target and scope:** removal or alteration of a target field, boundary qualifier, spectral premise, or nonclaim;
2. **Encoded algebra:** charge mismatch, response-null-vector failure, generator/field-coordinate type collapse, or broken normalization covariance;
3. **Dependency and provenance:** missing theorem IDs, source locators, hashes, or retained-boundary statements;
4. **Artifact synchronization:** stale generated tables, conflicting atlas twins, unresolved citations, or a manuscript/data mismatch;
5. **Claim boundary:** insertion of language that promotes a finite registry, validator, or empty file into a semantic completeness or exhaustion proof.

A passing suite establishes that these encoded corruptions are detected. It does not establish that every possible scientific error has been anticipated. The correct inference is

$$\text{declared corruption detected} \implies \text{certification gate functions as specified,}$$

not

$$\text{all tests pass} \implies \text{the scientific theorem is automatically true or exhaustive.}$$

G.3 Open scientific challenges

A same-target physical rival is not represented as a hostile file corruption. It must be presented as a mathematical or physical construction and judged against the theorem chain in [section 17](#). If it preserves all premises while changing an invariant entry of $\mathfrak{E}_{EW}$, it reopens the scientific result whether or not every artifact test passes.

Exact nonclaim boundary

The hostile-audit suite certifies the declared artifact boundary only. It neither proves semantic candidate completeness nor replaces specialist review of the electroweak arguments and imported QFT interfaces.

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